

Words Before Exit: CEO Linguistic Patterns and Forced Turnover Outcomes

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Abstract

We examine CEO language style during the final earnings conference calls before forced turnovers to identify speech patterns that precede dismissal and assess whether linguistic attributes can serve as predictive signals of forced turnover. Our findings reveal that CEOs nearing dismissal display distinct verbal communication patterns compared to their peers who retain their positions. Specifically, increased use of first-person pronouns, analytical language, power-related words, and a present-focused orientation is associated with a higher likelihood of forced turnover. In contrast, greater use of third-person pronouns, authentic language, affective words, and tentative language correlates with a lower likelihood of dismissal. These results remain robust when comparing fired CEOs to similarly situated retained CEOs using propensity score matching, entropy balancing, and difference-in-difference analyses. Furthermore, our analysis shows that these linguistic shifts are specific to CEOs, as CFOs do not display similar patterns. Additionally, we find that new CEOs following forced turnovers exhibit distinct communication styles compared to their ousted predecessors. Our study contributes to the literature on executive communication and corporate governance by demonstrating how CEO language can signal impending forced turnover, offering valuable insights into the role of verbal communication in corporate leadership transitions.

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1. Introduction

Deciding whether to retain or dismiss a CEO following poor performance is one of the most critical responsibilities of corporate boards. CEO departures can significantly impact a firm's strategy, operations, and overall performance, often heightening uncertainty among stakeholders (Jenter and Kanaan, 2015; Walsh and Seward, 1990). The transition to a new CEO further amplifies market uncertainty, which can only be mitigated through subsequent firm performance and disclosures (Bochkay et al., 2019). Research on CEO turnover suggests that corporate boards evaluate CEO quality based on firm performance and other indicators before making retention decisions (Jenter and Kanaan, 2015). One such indicator is the CEO's communication style, particularly during earnings conference calls, which serves as a crucial channel for signaling managerial competence to external stakeholders (Bochkay et al., 2019). In this study, we contribute to the CEO turnover literature by examining the verbal communication of CEOs during their final earnings conference call, identifying linguistic attributes that are associated with their eventual ousting and providing a novel perspective on the predictive power of language in forecasting executive tenure outcomes.

Communication researchers have shown that specific language features significantly influence how audiences perceive and evaluate both the speaker and the message itself (Miller et al., 2007). Listeners draw inferences about a speaker's attitude based on the language used during communication (Sperber and Wilson, 1987; Vonk, 1998, 2002; Pan et al., 2018). Previous studies in impression management highlight that top executives use various communication and disclosure strategies to shape stakeholder perceptions (e.g., Merkl-Davies and Brennan, 2007; Li, 2008; Graffin et al., 2011; Washburn and Bromiley, 2014). As König et al. (2018) note, "CEOs are under pressure not only to carefully select the details they wish to share with infomediaries but also to

package that information in a way that reduces complexity and subtly directs infomediaries' attention toward a positive interpretation.”

Quarterly earnings conference calls are among the most important forms of disclosure for external stakeholders and capital markets (Frankel et al., 1999; Kimbrough, 2005). These dynamic, less structured events offer valuable insights compared to static, formal disclosures such as the annual 10-K filing (Frankel et al., 1999; Blau et al., 2015). Given the nature of these calls, CEOs have considerable discretion in shaping the information presented to influence stakeholders (Matsumoto et al., 2011; Pan et al., 2018). For instance, managers often strategically externalize poor performance by attributing it to market or industry conditions through blame statements in their speech during earnings conference calls (Noh and Zhou, 2022). Recent research has explored the impact of CEO verbal communication on stock performance, analyst assessments, and investor opinions (Pan et al., 2018; Green et al., 2019; Guo et al., 2021; Noh and Zhou, 2022; Pollock et al., 2023). However, the extent to which CEOs alter their language style just before being ousted remains underexplored. This gap is crucial because if words can indeed influence markets, it raises an important question: Do CEOs communicate differently just before they are fired compared to their counterparts who remain in position?

CEOs' language style is reflected in the types and frequencies of words they use (Pan et al., 2018). Theories of language analysis suggest that word frequencies can indicate the degree of attention to specific domains (Boyd and Schwartz, 2021). For example, pronouns serve as indicators of the focus of attention rather than just grammatical elements. Similarly, emotion words do not directly convey the experience of emotions; instead, they function as tools for signaling attention to affective states. A growing body of literature uses textual analysis to examine the linguistic choices executives make when communicating firm performance and news (Henry,

2006; Tetlock et al., 2008; Mayew and Venkatachalam, 2012; Price et al., 2012; De Amicis et al., 2021). Language style not only mirrors the speaker's mood and thoughts but also shapes how the audience interprets the message. For instance, mood congruence or consistency can enhance the persuasiveness of information (Mayer et al., 1992; DeSteno et al., 2004). Given that language plays a crucial role in forming audience impressions, it significantly influences both the quality and quantity of message delivery (Larrimore et al., 2011; Toma and D'Angelo, 2015). Our study builds on prior research by applying textual analysis to CEO verbal communication, focusing on linguistic elements such as personal pronouns, affective and cognitive language, and temporal orientations.

We examine a sample of 45,438 CEO-quarter conference call observations, including 224 unique forced turnovers, between 2010 and 2018. As earnings conference calls serve as crucial communication channels for CEOs to signal their ability and competence to the market and corporate board (Pan et al., 2018; Bochkay et al., 2019; Green et al., 2019; Noh and Zhou, 2022), we argue that CEOs strategically select their language to influence stakeholders' perceptions prior to their dismissal. In other words, we contend that these CEOs are likely aware of their impending departure, and this awareness is reflected in their verbal communication during their final conference calls. Using textual analysis, we measure a broad range of linguistic attributes in conference call transcripts, such as the use of personal pronouns, analytical thinking, authenticity, affective and power-related words, and temporal focus. Our analyses reveal that specific language attributes in conference calls are significantly associated with forced CEO turnover. Notably, the use of more analytical language, frequent first-person singular pronouns, power-related words, and a present-focused orientation is significantly associated with a higher likelihood of CEO dismissal.

In contrast, a greater use of authentic language, third-person pronouns, affective words, and tentative language is linked to a lower likelihood of forced turnover.

The results remain robust when using both propensity score matching (PSM) and entropy balancing methods to address potential selection bias and endogeneity concerns. We match firms with forced CEO turnover to those without CEO turnover, ensuring that they share similar characteristics. By pairing each CEO's conference call prior to dismissal with a call from a CEO who retained their position—based on firm-level characteristics—we achieve comparable firm performance between the two groups. Thus, the PSM and entropy balancing approaches enable us to examine differences in linguistic measures, showing that fired CEOs tend to use more analytical language, more first-person pronouns, and fewer affective and tentative words compared to CEOs who remained in their roles.

Furthermore, we investigate the verbal communication of new CEOs during their first earnings conference call following both forced and voluntary turnovers. Using a difference-in-difference analysis, our findings show that new CEOs after a forced turnover significantly alter their speech patterns during these initial calls, compared to the changes observed in new CEOs following a voluntary turnover. Additionally, we examine CFOs' verbal communication to determine whether these linguistic patterns are unique to CEOs. The results indicate that the shifts in communication style before forced turnovers are specific to the CEO role, as CFOs do not exhibit similar linguistic changes.

Our paper provides valuable insights into the predictive power of CEO language styles in anticipating forced turnovers, highlighting how a CEO's choice of words during conference calls can signal their impending dismissal. These findings contribute to a deeper understanding of the

dynamics of CEO verbal communication during conference calls and its implications for forced turnover.

This study advances our understanding of CEO language style by providing a comprehensive analysis of how linguistic attributes in CEOs' verbal communication are linked to their forced turnover. Our study, thus, adds to the growing body of literature on the determinants of forced CEO turnovers and the information signals communicated to the market and corporate boards (Campbell et al., 2011; Jenter and Kanaan, 2015; Dasgupta et al., 2018; Jenter and Lewellen, 2021). Additionally, we contribute to the literature on CEO communication during voluntary disclosures, such as earnings conference calls, and their impact on stakeholders and the market (Matsumoto et al., 2011; Price et al., 2012; Pan et al., 2015; Blau et al., 2015; Bochkay et al., 2019). We emphasize the significance of conference calls as dynamic information channels for external stakeholders by showing that specific linguistic features in CEOs' speech during their final earnings call are predictive of their dismissal. These findings provide valuable insights into how CEO language style influences career outcomes, revealing that CEOs facing imminent dismissal strategically adjust their language, and that these adjustments can serve as early indicators of forced turnover.

2. Literature review

There is an extensive body of literature on corporate disclosure that examines how firm performance influences managerial communication and disclosure practices. Prior research has explored how executives adjust their disclosures in response to performance outcomes (Li, 2008; Brown and Tucker, 2011) and how managers' personal characteristics shape the nature and tone of these communications (Bertrand and Schoar, 2003; Davis et al., 2015). These studies generally

operate under the premise that market demand for information stems from uncertainty surrounding firm performance, paired with managers' private insights and strategic intent. Since CEOs are central to the formulation and communication of firm policies, strategies, and disclosures, any uncertainty about their capabilities amplifies the market's need for information to reduce ambiguity and risk. Supporting this notion, Bochkay et al. (2019) find that CEOs' forward-looking statements in conference calls tend to decline as their tenure progresses and the market gains a clearer understanding of their abilities.

While the existing literature has deepened our understanding of how executives, particularly CEOs, adjust disclosure practices in response to firm performance and managerial characteristics, less is known about how CEOs communicate in high-stakes situations, such as in the period leading up to a forced departure. Our study addresses this gap by integrating insights from research on corporate communication, CEO turnover, and earnings conference calls. Specifically, we investigate whether CEOs exhibit distinctive language patterns in their final conference call before dismissal. By doing so, we contribute to a more nuanced understanding of CEO discourse under conditions of personal and organizational uncertainty. To guide this analysis, we structure our literature review into three key sections that build on these intersecting research streams.

2.1 Literature on language and communication

Previous research has demonstrated that language possesses a distinctive ability to reveal key aspects of an individual's personality, cognitive processes, and thinking patterns (Tausczik and Pennebaker, 2010). Studies in communication have shown that specific linguistic features embedded within messages significantly shape audience perceptions and evaluations of both the speaker and the content delivered (Miller et al., 2007). In the corporate context, CEOs' spoken and

written language serves as a reflection of their personalities, offering valuable insights into their emotional states, cultural backgrounds, and political views (Chatterjee and Hambrick, 2007; Burgoon et al., 2016). As such, linguistic analysis has emerged as a practical and reliable approach to understanding executive behaviors and decision-making processes (Pennebaker et al., 2003; Markowitz, 2018; Akstinaite et al., 2020).

The literature on corporate communication further highlights that language plays a crucial role in shaping a firm's ability to allocate resources efficiently, as CEOs' linguistic choices influence stakeholder uncertainty, investment decisions, and broader market perceptions (Hales et al., 2011; Guo et al., 2017; König et al., 2018; Pan et al., 2018; Huang et al., 2021; Pollock et al., 2023). This is particularly relevant during moments of organizational stress or uncertainty, when CEO communication becomes even more pivotal.

Aligned with this, the impression management literature examines how executives strategically employ communication tactics to influence market and stakeholder perceptions (Pan et al., 2018). For instance, prior studies have explored how managers selectively choose media outlets to disseminate positive news (Washburn and Bromiley, 2014), manipulate the timing of news releases to minimize the impact of negative events (Graffin et al., 2011), and craft specific statements to present firm operations and prospects in a favorable light (Pan et al., 2018). Discretionary disclosure strategies serve as a means for executives to justify firm actions and reduce external skepticism through deliberate content framing and narrative construction (Merkl-Davies and Brennan, 2007). Moreover, CEO language can reveal personality traits such as hubris, which, when left unchecked, may result in destructive behaviors and suboptimal corporate outcomes (Craig and Amernic, 2011; Sadler-Smith et al., 2019). Recognizing these linguistic

markers provides an important framework for mitigating risks associated with hubristic leadership (Akstinaite et al., 2020).

Taken together, these streams of research suggest that CEO language is not only reflective of internal psychological and strategic factors but also a deliberate tool to manage impressions and shape external narratives.

2.2 Literature on CEO turnover

Previous literature has established a negative relationship between CEO turnover and stock returns, with poor firm performance increasing the likelihood of executive dismissal (Murphy and Zimmerman, 1993). Jenter and Kanaan (2015) further demonstrate that negative industry shocks can also trigger managerial turnover, although firm-specific shocks tend to have a more pronounced effect. Boards of directors, however, are generally cautious in dismissing CEOs due to the disruptive nature of such decisions and the uncertainty surrounding the selection of a capable successor (Duchin and Schmidt, 2013). In this context, periodic CEO disclosures play a critical role in shaping labor market perceptions of CEO competence, which in turn can influence the probability of dismissal (Hermalin and Weisbach, 1998; Jenter and Lewellen, 2021). As a result, CEOs have a strong incentive to manage market perceptions by framing their actions and strategies in a favorable light (Noh and Zhou, 2022).

In a recent study, Noh and Zhou (2022) find that when a board accepts a CEO's attempt to externalize blame for poor performance, the likelihood of dismissal is reduced. This finding highlights the power of CEOs' strategic use of blame in their communication, which serves as an effective tool for influencing both market beliefs and external stakeholder perceptions. Furthermore, CEO turnover and the appointment of a new CEO often lead to increased market

uncertainty regarding a firm's future performance, as such transitions are frequently accompanied by strategic changes. This uncertainty can only be mitigated by closely monitoring firm performance and analyzing the CEO's communication style in subsequent periodic disclosures (Bochkay et al., 2019).

On the whole, while CEO turnover is generally associated with negative market reactions, strategic communication by the CEO—such as managing blame and shaping perceptions—can influence the board's and stakeholders' views on leadership effectiveness, confidence in the CEO's ability to navigate challenges, and overall managerial competence, potentially reducing the likelihood of dismissal. Our understanding of these dynamics can be further enriched by examining how CEO language evolves in the final stages of tenure, particularly before forced turnover.

2.3 Literature on earnings conference calls

Earnings conference calls are a vital form of voluntary disclosure to external stakeholders and capital markets (Frankel et al., 1999; Kimbrough, 2005). These calls typically consist of two segments: the management discussion (MD), where quarterly (or annual) results are presented, and the question-and-answer (Q&A) session, where financial analysts engage with management. Unlike formal documents such as annual reports, press releases, and news articles, which are static and highly scripted (Blau et al., 2015), conference calls provide a dynamic, less structured environment. This format allows executives to reveal their personal communication style more spontaneously, offering valuable insights that market participants and stakeholders can analyze (Price et al., 2012; De Amicis et al., 2021).

Much of the research on conference calls has focused on qualitative characteristics of the information conveyed, such as managers' tone (Price et al., 2012) and linguistic complexity (Bushee et al., 2011). Earnings call transcripts have also been used to assess managerial knowledge

(Li et al., 2014), disclosure horizon (Brochet et al., 2015), and spontaneity (Lee, 2016). Recent studies have expanded this body of literature by examining how CEOs' disclosure practices evolve over their tenure and how these practices influence investor perceptions and stock returns (Noh and Zhou, 2022; Rennekamp et al., 2022).

CEOs use these calls not only to present earnings but also to clarify and promote investor reactions to the disclosed information (Hollander et al., 2010). Previous studies suggest that managers often use concrete language to justify firm performance, which can positively influence investor sentiment (Pan et al., 2018). Furthermore, attributes such as the newness, simplicity, and unscripted nature of CEO verbal communication are associated with greater investor consensus (Guo et al., 2021). CEO personality traits, such as extraversion, have been linked to enhanced investor recognition and favorable firm outcomes (Green et al., 2019). In a similar vein, the language style of high-profile, or “celebrity” CEOs—characterized by positivity, concreteness, certainty, and self-regard—has a notable impact on stakeholder perceptions (Green et al., 2019; Pollock et al., 2023).

Overall, previous studies highlight the significant role of CEO communication, particularly in the context of earnings conference calls, CEO turnover, and corporate disclosure practices. Research consistently shows that the language used by CEOs offers valuable insights into their personalities, cognitive processes, and strategic intent, which can profoundly influence investor perceptions and market outcomes. CEOs strategically manage their communication to shape stakeholder beliefs, justify firm performance, and mitigate uncertainty, especially during periods of executive turnover or organizational transition. The dynamics of language in earnings calls, from tone to linguistic complexity, reflect both individual leadership traits and broader corporate strategies. This body of research sets the stage for our study, which aims to deepen our

understanding of how CEO language shifts in the final stages of their tenure, particularly leading up to forced turnover, and how these linguistic shifts may signal the impending dismissal of a CEO. By integrating insights from the fields of corporate communication, CEO turnover, and disclosure practices, our study offers a comprehensive approach to understanding the intersection of leadership, language, and market reactions.

3. Data and methodology

In this section, we describe our data and outline the methodology we use to examine the linguistic patterns of CEOs during earnings conference calls and their association with forced turnover. We first describe the data collection process, including the identification of CEO turnover events and the compilation of firm-level financial data in Section 3.1. Then, we discuss the use of Linguistic Inquiry and Word Count (LIWC) software to analyze various linguistic attributes in CEO speech, focusing on standard linguistic dimensions, psychological processes, and cognitive traits in Section 3.2. In Section 3.3, we present summary statistics for firm and CEO characteristics.

3.1 Data

We begin our sample collection by sourcing data from the ExecuComp database, covering the period from 2010 to 2018. The sample starts in 2010, following the financial crisis, to avoid including turnovers influenced by recession-related factors. We end the sample in 2018 to exclude the economic volatility and potential forced turnovers (and associated CEO appointments) linked to the COVID-19 period. We identify CEO turnovers by tracking changes in the CEO flag within ExecuComp. To classify turnover events, we hand-collect news reports from Factiva and examine proxy statements to determine whether each departing CEO left voluntarily or was forced out, as

forced turnovers are often disguised as retirements. We exclude interim CEOs from the sample and categorize each turnover event as either forced or voluntary based on the available information from news announcements.

Following the classification scheme of Parrino (1997) and Campbell et al. (2011), a turnover is classified as forced if it meets one or more of the following criteria: (1) the departure was not announced at least six months in advance; (2) the departing CEO is under the age of 60 and does not leave due to health issues, death, or to accept another position elsewhere; (3) the departing CEO does not remain on the firm's board of directors after stepping down; or (4) the CEO retires or resigns under questionable circumstances, or the company explicitly fires, demotes, or forces the CEO out (e.g., due to policy disagreements). Panel A of Table 1 presents the total number of CEO turnover events, including both overall turnovers and forced turnovers, for each year in the sample. As shown, we identify 1,706 CEO turnovers out of 18,131 firm-year observations in ExecuComp, of which 347 (20.3%) are classified as forced. The remaining 1,359 are classified as voluntary turnovers (e.g., due to retirement).

Next, we obtain quarterly earnings conference call transcripts from Capital IQ. Firm-level financial data is collected from several databases, including Compustat, the Center for Research in Security Prices (CRSP), the Institutional Brokers Estimate System (I/B/E/S), and Thomson Reuters. The sample selection process, after merging all data sources, is summarized in Panel B of Table 1. Panel B outlines the step-by-step refinement of our dataset to ensure comprehensive coverage of quarterly CEO turnover events, with a particular focus on forced turnovers. We begin with 1,706 CEO turnover observations from ExecuComp, which we convert to a quarterly format and merge with firms that have complete data in Compustat and CRSP, yielding 69,087 observations, including 319 forced turnovers. Finally, adding earnings conference call transcripts

from Capital IQ results in a final dataset comprising 45,438 CEO-quarter observations, including 224 cases of forced turnovers and 923 cases of voluntary turnovers.

3.2 Textual analysis of CEO language

There is a growing body of literature in finance analyzing various forms of textual data, beginning with the work of Tetlock (2007), with many studies focusing on corporate disclosures and linguistic attributes. For example, prior research examines company disclosures to measure tone (Tetlock et al., 2008; Loughran and McDonald, 2011), uncertainty (Loughran and McDonald, 2013), and readability (Li, 2008). Textual analysis has become a widely used tool among scholars for conducting content analysis. Words play a critical role in shaping audience perceptions, and even subtle variations in word choice can reveal important insights into a speaker's motives and beliefs (Berry et al., 1997; Semin, 2008; Pan et al., 2018). Notably, two messages conveying similar content can vary significantly in their effectiveness depending on the specific words used (Pennebaker and King, 1999). For instance, the use of function words—such as pronouns, articles, and conjunctions—is closely linked to how people express their social and psychological worlds, and the ability to use these words effectively in speech reflects basic social skills (Tausczik and Pennebaker, 2010).

Linguistic attributes serve as “honest signals” of a speaker's intentions, reflecting their motivations, beliefs, and attitudes (Pentland, 2008). For instance, personal pronouns reveal attentional focus (Rude et al., 2004), while verb tenses offer insights into how individuals cognitively process events (Pasupathi, 2007). People tend to use the past tense more often when discussing disclosed or resolved situations and the present tense when referring to ongoing or undisclosed events (Pasupathi, 2007; Tausczik and Pennebaker, 2010). Both pronoun use and verb

tense choices help pinpoint the focus and orientation of speech, shedding light on speakers' priorities and intentions (Kacewicz et al., 2014; Burgoon et al., 2016). Additionally, individuals who experience uncertainty often rely on more tentative language and filler words (Tausczik and Pennebaker, 2010). Prior linguistic research also shows that verbs and quantitative terms are typically more concrete and precise than adjectives or vague quantifiers, which are commonly used when people generalize (Semin et al., 2005; Larrimore et al., 2011). Collectively, these studies highlight that an individual's cognition, motives, and behaviors are shaped by their circumstances, and their communication style offers meaningful clues about their psychological state and situational context.

We measure the linguistic attributes of earnings conference calls using the Linguistic Inquiry and Word Count (LIWC) software. LIWC is a word-counting tool equipped with validated dictionaries that have demonstrated both internal reliability and external validity (Pennebaker and Chung, 2012). The software analyzes conference call transcripts by matching each word in the text to entries in its dictionary and calculating the percentage of words that fall into each predefined category. For each transcript, LIWC generates approximately eighty output variables. These include summary language metrics (such as analytical thinking, clout, authenticity, and emotional tone), general text descriptors (e.g., words per sentence, the percentage of target words captured by the dictionary, and the proportion of words longer than six letters), standard linguistic dimensions (e.g., the percentage of pronouns, articles, verbs), categories related to psychological constructs (e.g., affective and cognitive processes, drives), personal concern categories (e.g., work, home, leisure), informal language markers (e.g., assents, fillers, swear words, netspeak), time orientation (past, present, and future), and punctuation usage (e.g., periods, commas).

We focus on the linguistic measures extracted by LIWC that capture standard linguistic dimensions, as well as psychological and cognitive processes, all of which have been explored across various strands of prior literature. Panel A of Table 2 reports summary statistics for these linguistic attributes, alongside firm-level and CEO-level variables, for two groups: CEOs who remained in their positions (“No Turnover,” using all conference calls not followed by a turnover) in Columns (1) to (3), and CEOs who were forced out (“Forced Turnover,” using only the last conference call before dismissal) in Columns (4) to (6). Columns (7) and (8) present the differences in linguistic measures between the two groups. We find notable differences in linguistic attributes between dismissed and retained CEOs. For instance, the average use of the pronoun “I” is 0.700 in the forced turnover sample, compared to 0.613 in the no-turnover sample—a statistically significant difference at the 1% level. Similarly, the use of tentative language averages 1.17 for retained CEOs and 1.08 for dismissed CEOs, with this difference being statistically significant at the 5% level.

Additionally, we include the mean and standard deviations of these linguistic measures from the LIWC manual (the last two columns) to illustrate how CEO language differs from that of the general population. Notably, CEOs exhibit higher levels of analytical thinking, with an average analytic score of 91.31 for No Turnover CEOs and 91.59 for Forced Turnover CEOs, compared to LIWC’s grand mean of 56.34. This suggests that CEOs, regardless of turnover status, tend to adopt a distinct communication style, setting them apart from the average speaker captured by LIWC benchmarks.

As discussed in Section 2, earnings conference calls provide CEOs with a platform to strategically shape perceptions of their leadership and firm performance. The variations in communication style observed in Table 2 may reflect deeper psychological and cognitive

dynamics, potentially signaling CEOs' awareness of their risk of dismissal. In our analysis, we examine these linguistic differences to assess their relationship with forced turnovers.

3.3 Firm and CEO characteristics

Panel B of Table 2 presents firm and CEO characteristics for those who experienced forced turnover compared to those who remained in their positions. Several notable differences emerge between the two groups. Firms with forced CEO turnover tend to have significantly lower return on assets (ROA) (0.002 vs. 0.011, p -value = 0.000) and higher reported losses (0.129 vs. 0.083, p -value = 0.012), indicating poorer financial performance preceding dismissal. Additionally, firms with forced turnovers have a lower market-to-book ratio (MTB) (1.761 vs. 2.004, p = 0.007), suggesting that investors may perceive them as having lower growth potential. CEO tenure (measured as the natural logarithm of the number of years between the CEO's appointment date and the current period's end date) is slightly lower for those who are forced out (1.736 vs. 1.862, p = 0.050), which may imply that recently appointed CEOs are at a higher risk of dismissal. Moreover, ownership stakes are significantly lower for dismissed CEOs (0.008 vs. 0.021, p = 0.000), possibly reflecting weaker alignment with shareholder interests. Age (measured as the natural logarithm of the CEO's age) also differs slightly, with dismissed CEOs being slightly younger on average (3.991 vs. 4.043, p = 0.000). The gender composition of forced turnover cases shows a lower proportion of female CEOs (0.068 vs. 0.047, p = 0.000). In contrast, leverage, analyst following, institutional ownership, and earnings surprise do not show statistically significant differences between the two groups. These findings highlight key firm- and CEO-level characteristics associated with forced turnover, particularly in terms of financial underperformance, tenure, ownership, and demographic attributes.

4. Choosing linguistic measures of forced CEO turnover

Language conveys crucial information about our identity, thoughts, emotions, and psychological states (Pennebaker, 2011). Therefore, it is reasonable to assume that language style varies across different contexts, revealing distinct verbal patterns. In this study, we explore the linguistic features of CEOs during earnings conference calls and their association with forced turnover.

To guide our analysis, we select word categories based on a comprehensive review of prior research. Various strands of literature have examined linguistic markers related to deceptive language, persuasive communication, and cognitive emotions in speakers' language styles. Previous studies in the context of financial reporting have demonstrated how language can signal deceptive behavior. For example, Larcker and Zakolyukina (2012) analyzed linguistic cues in CEO and CFO statements during earnings conference calls to predict deceptive financial reporting, such as auditor changes and late filings. In our study, we apply linguistic markers from this literature to examine whether CEOs use strategic language to project confidence, authority, focus, control over narratives, and to shift blame or alter temporal framing in their conference calls before being forced out.

Researchers have explored the use of "I" pronouns across various strands of literature, yielding mixed findings in the context of deception. Some studies suggest that liars tend to use fewer self-references (Newman et al., 2003; Larcker and Zakolyukina, 2012), while others highlight the ambiguous relationship between linguistic measures and deception, which can vary depending on the theoretical framework (Hauch et al., 2015; Markowitz and Hancock, 2019). In negative situations, lies often involve fewer first-person singular pronouns, potentially as a psychological distancing mechanism (Newman et al., 2003; Hauch et al., 2015). However, in

neutral contexts, lies may feature more frequent use of first-person singular pronouns, perhaps to enhance the speaker's credibility (Pennebaker, 2011; Larcker and Zakolyukina, 2012; Hauch et al., 2015). In this study, we examine the emotional and affective content of earnings conference calls to better understand the use of self-referential pronouns. Specifically, we focus on the "I" pronoun and words related to the "affective process" from LIWC's linguistic categories to assess whether CEOs display signs of deceitful behavior before being forced out.

The degree to which a speech is well-rehearsed is another key linguistic indicator explored in deception literature (Larcker and Zakolyukina, 2012; Vrij, 2014). When speakers communicate authentically or honestly, they tend to speak more spontaneously, with less premeditation. Authenticity also reflects the extent to which a person self-monitors their speech (Newman et al., 2003; Kalichman and Smyth, 2023; Markowitz et al., 2023). Therefore, we include the "authentic" category from LIWC to examine its relationship with forced CEO turnover. Additionally, prior literature suggests that the use of tentative language may signal psychological distancing between the speaker and their statements (Newman et al., 2003; Larcker and Zakolyukina, 2012). As such, we also incorporate the "tentative" category from LIWC in our analysis of linguistic markers to further investigate the connection between language and forced turnover.

Speakers may craft persuasive arguments or use specific language to appear more convincing when making deceptive claims. Pan et al. (2018) discuss how CEOs' use of concrete language enhances the persuasiveness of their messages and generates positive investor reactions. Another study demonstrates that a higher level of analytical thinking is positively associated with message persuasiveness (Ta et al., 2022). Speakers who exhibit greater analytical thinking tend to use words that reflect formal, logical thought patterns, while those with less analytical thinking often rely on more intuitive and personal language (Pennebaker et al., 2014; Jordan et al., 2019).

Consequently, we examine the “analytic” category in LIWC, which captures the analytical thinking of CEOs as previous research has shown that this linguistic marker is associated with the persuasiveness of conference calls (Ta et al., 2022).

Temporal focus reflects a speaker’s attention to the past, present, and future (Bluedorn and Wright, 2003; Shipp et al., 2009; Gamache and McNamara, 2019). Gamache and McNamara (2019) discuss how a CEO’s temporal focus shapes their perspective and influences their sensitivity to media coverage during major acquisitions. A present focus, in particular, signals an orientation toward the current moment—what is happening “here and now” (Nadkarni and Chen, 2014). Given that our study examines CEO speech patterns during their final conference call before being ousted, understanding their present focus is crucial, as it reflects their attention to the immediate situation. Therefore, we incorporate the “present focus” category from LIWC to assess its association with forced turnover.

In the literature on linguistic psychological processes, Lerner and Keltner (2000) explore how specific emotions influence judgment and decision-making. For example, the use of the pronoun “they” is linked to greater externalization in situations that evoke anger. To assess the “affective process” in CEO verbal communication, we examine the frequency of the pronoun “they” as a measure of externalization. In the context of the final conference call before forced turnover, the use of “they” reflects how CEOs externalize responsibility or blame. We compare this with the “tentative” measure, which signals psychological distance from their statements.

A study on CEO hubris finds that the use of power-related words is an indicator of excessive CEO confidence (Akstinaite et al., 2020). Language, especially in the context of deceptive behavior and persuasive communication, is a valuable tool for analyzing psychological processes and personality traits (Garrard et al., 2014). One key indicator of hubris identified by

Garrard et al. (2014) is the use of power expressions. Therefore, we analyze the use of power-related words from LIWC to assess the level of CEO confidence during the final conference calls before their dismissal.

5. Empirical tests and results

In this section, we present the empirical tests and results that investigate the relationship between CEO language style and forced turnover. We begin by conducting panel data regressions in Section 5.1, where we explore the association between various linguistic attributes in CEOs’ verbal communication and the likelihood of forced turnover. This approach allows us to account for firm-level characteristics and potential confounding variables. In Section 5.2, we address concerns related to selection bias and endogeneity by employing propensity score matching (PSM) and entropy balancing techniques. These methods enable us to compare firms with and without forced turnovers while ensuring that the matched samples exhibit similar characteristics, thereby providing more robust and reliable results. Together, these empirical strategies offer comprehensive insights into the predictive power of CEO language style in anticipating forced turnover.

5.1 Panel data regressions

In this section, we investigate the relationship between CEO linguistic measures extracted from LIWC and the likelihood of forced turnover. To do so, we employ a logistic regression model. Our dependent variable, *ForcedTurnover*, is binary—taking a value of one if the CEO is forced out and zero if there is no turnover. The key linguistic measures included in our analysis are “analytic,” “authentic,” “I” pronoun, “they” pronoun, “affective process,” “tentative,” “power,”

and “focus present,” as outlined in the previous section. Our primary analysis focuses on linguistic features extracted from the management discussion (MD) section of earnings conference calls. For robustness, we also examine the relationship using linguistic measures from the question-and-answer (Q&A) section of the calls.

Consistent with prior studies (e.g., Pan et al., 2018; Bochkay et al., 2019; De Amicis et al., 2021; Comprix et al., 2022; Guo and Zhong, 2023), we control for firm-level characteristics that account for differences in firms’ information environments, operations, performance, and uncertainty. These control variables include *Firm Size* (measured as the logarithm of total assets), *ROA* (return on assets, calculated as earnings before extraordinary items scaled by total assets), *MTB* (market-to-book ratio, defined as the market value of equity plus the book value of liabilities divided by the book value of assets), *Return* (average monthly stock return over the quarter), *Leverage* (measured as the ratio of long-term debt to total assets), and *Loss* (an indicator variable equal to one if EPS is negative). Additionally, we include *Earnings Surprise* (measured as the difference per share between actual earnings and the mean consensus analyst forecast, scaled by the stock price at the end of the period), *Following* (analyst following, measured as the logarithm of the number of analysts issuing an earnings forecast), and *Institutional Ownership* (the percentage of institutional holdings, derived from 13-F data, measured as the number of shares held by institutions divided by the total number of shares outstanding).

Furthermore, we control for CEO characteristics, including *Tenure* (measured as the natural logarithm of the number of years between the CEO’s appointment date and the current period’s end date), *Age* (measured as the natural logarithm of the CEO’s age), *Gender* (a dummy variable equal to one if the CEO is female and zero if male), and *Ownership* (measured as the percentage of shares owned by the CEO). These controls align with prior research (e.g., Pukthuanthong et al.,

2018; Green et al., 2019; Noh and Zhou, 2022). All these variables are also defined in the Appendix.

We start by testing whether these CEO attributes provide any indication that a dismissal is approaching by estimating the following regression, which compares forced turnovers to cases with no turnover:

$$\begin{aligned}
ForcedTurnover_{it} = & \beta_0 + \beta_i LIWCVar_{it} + \alpha_1 FirmSize_{it} + \alpha_2 ROA_{it} + \alpha_3 MTB_{it} + \\
& \alpha_4 Leverage_{it} + \alpha_5 Return_{it} + \alpha_6 Earning Surprise_{it} + \alpha_7 Loss_{it} + \\
& \alpha_8 Following_{it} + \alpha_9 Intsown_{it} + \alpha_{10} CEOTenure_{it} + \alpha_{11} CEOAge_{it} + \\
& \alpha_{12} CEOGender_{it} + \alpha_{13} CEOown_{it} + \delta_1 Time FE_t + \delta_2 (Firm FE_i \text{ or } Industry FE_i) + \\
& \varepsilon_{it}
\end{aligned} \tag{1}$$

where *ForcedTurnover_{it}* is an indicator variable equal to one if there is a forced turnover between the current and the next conference call, and zero if no turnover occurs during that interval. *LIWCVar_{it}* is linguistic measures extracted from conference calls in each quarter. Firm and CEO level variables are measured as of the end of the quarter.

Table 3 presents the regression results for the model specified in Equation (1). We estimate Models (1) through (4) by incorporating CEO speech measures and control variables while varying the specifications for fixed effects to control for unobserved time, firm, or industry fixed effects. Specifically, Column (1) reports a logit regression of forced CEO turnover with firm fixed effects and year-quarter fixed effects.¹ Column (2) replaces firm fixed effects with fiscal year and industry

¹ Since Model (1) uses a logistic regression with firm fixed effects, the sample size is considerably smaller. This is because firms with no forced turnovers during the sample period are automatically excluded due to the lack of variation in the dependent variable. To address this limitation, we re-estimate Model (1) of Table 3 and the same model of related tables using a linear OLS model, which retains more observations. The OLS results are generally consistent with those from Model (1) of Table 3 and the other tables. For example, in untabulated results, we find that all linguistic variables—except *Tentative*—remain statistically significant and retain their expected signs.

fixed effects. Column (3) includes both industry and year-quarter fixed effects, while Column (4) applies industry and quarter fixed effects.

The results in Table 3 indicate that CEO linguistic variables are significantly associated with forced turnover, suggesting that CEOs exhibit distinct language patterns during their final conference calls before dismissal. Across all models, linguistic features such as “analytic,” “I” pronoun, “power,” and “focus present” are positively and significantly associated with the likelihood of forced turnover. Conversely, variables such as “authentic,” “they” pronoun, “affective process,” and “tentative” are negatively related to the probability of forced turnover.

For instance, in Model (2), a one-percent increase in the usage of “I” pronoun increases the log-odds of forced turnover by 0.399, corresponding to an approximate 49.08% increase in the odds of forced turnover. This effect is statistically significant at the 5% level ($p\text{-value} = 0.012$). Conversely, a one-percent increase in the usage of “they” pronoun decreases the log-odds of forced turnover by 0.721, or an approximate 51.40% reduction in the odds of forced turnover, also statistically significant at the 5% level ($p\text{-value} = 0.022$).

These results offer important insights into the predictive power of speech variables from CEOs’ final conference calls in forecasting forced turnover. Our findings align with prior literature on deception, persuasion, cognitive emotions, and psychological processes, reinforcing the rationale for selecting specific linguistic features. Notably, the increased use of first-person singular pronouns suggests a heightened self-focus, while the reduced use of affective words indicates a more neutral speech pattern—both commonly associated with deceptive behavior. These findings are consistent with previous research on the role of emotional content in detecting deception (Pennebaker, 2011; Larcker and Zakolyukina, 2012; Hauch et al., 2015).

Furthermore, in line with recent studies (Kalichman and Smyth, 2023; Markowitz et al., 2023), we find that authenticity is negatively associated with forced CEO turnover. A lower authenticity measure further supports the presence of deceptive behavior, suggesting that, during their final conference call, departing CEOs exhibit more carefully managed communication and heightened self-monitoring.

In their well-rehearsed speeches preceding dismissal, CEOs tend to use more analytical language, likely to enhance the persuasiveness of their statements. This finding aligns with the work of Pennebaker et al. (2014) and Ta et al. (2022), which link analytical language to more persuasive message delivery. This persuasive effort is reflected in a reduced psychological distance from their statements, as evidenced by the lower frequency of tentative words. Similar to Lerner and Keltner (2000), we observe a negative coefficient on the “they” pronoun, suggesting a decreased tendency to externalize responsibility is associated with a higher likelihood of dismissal, indicating attributing outcomes by a stronger self-focus rather than external factors. Additionally, consistent with Akstinaite et al. (2020) on CEO hubris, the increased use of power-related words signals heightened confidence in their verbal communication. A stronger focus on the present further suggests that these CEOs are possibly well aware of their current circumstances, emphasizing the immediate “here and now.” Collectively, these findings suggest that language used during critical organizational events carries meaningful signals about potential changes in executive leadership.

Our analysis of firm-level controls shows that return on assets (ROA), stock return, and the market-to-book ratio (Models 1 and 3) are negatively and significantly associated with the likelihood of forced CEO turnover. This suggests that larger firms, as well as those with higher profitability, better stock performance, and a stronger market valuation, are less likely to

experience CEO dismissals. Conversely, leverage exhibits a significant positive relationship with forced turnover, indicating that CEOs of highly leveraged firms face a greater likelihood of dismissal, likely due to the increased risk of financial distress. However, factors such as earnings surprise, loss measures, the number of analysts following the firm, and the percentage of institutional ownership do not appear to have a significant impact on forced CEO turnovers. Regarding CEO characteristics, those CEOs longer tenures and female CEOs are more likely to be dismissed, whereas CEOs with larger ownership stakes and older CEOs face a lower likelihood of dismissal.

Next, we analyze the linguistic features of CEO conference calls leading up to forced turnovers, focusing on the last four conference calls prior to dismissal. This analysis provides insights into whether CEOs gradually adjust their language in anticipation of potential dismissal, suggesting that they may become aware of their impending exit—or at least recognize the possibility—well before it occurs. The results of the logistic regressions for the trend analysis are presented in Table 4 by following the same CEOs with a forced turnover over four consecutive conference calls before they are dismissed. Table 4 consists of four panels, each corresponding to one of the four conference calls preceding a forced turnover. The panels reveal a clear trend in language changes across the fourth (-4), third (-3), second (-2), and first (-1) conference calls before forced CEO turnovers. In the fourth conference call, nearly a year before the forced turnover (Panel A), no significant linguistic shifts are observed, indicating stability in language use. However, from the third conference call onward (Panel B), subtle changes begin to appear, particularly in variables such as affective processes and present focus, though these shifts are still modest. By the second conference call (Panel C), these changes become more pronounced, with certain linguistic dimensions showing statistically significant variations. This suggests that CEOs’

communication becomes increasingly sensitive as forced turnover approaches. The most dramatic shifts are evident in the final conference call (-1) (Panel D), where linguistic variables such as analytic thinking, pronoun use, and expressions of authenticity show highly significant changes.² These findings suggest that CEOs adjust their communication style in meaningful ways as their forced turnover becomes imminent, with the magnitude of change increasing progressively closer to their dismissal.

Next, we examine voluntary turnovers in a manner similar to our analysis of forced turnovers in Table 3, to assess whether CEOs exhibit specific linguistic patterns prior to voluntary departures. Table 5 presents the logistic regression results for voluntary CEO turnover, with the dependent variable being binary: equal to 1 if the CEO turnover was voluntary and 0 if there was no turnover. Both tables assess the predictive power of linguistic features from CEOs' earnings call discussions, focusing specifically on the management discussion (MD) section. While the primary focus of this study is on forced turnover, Table 5 highlights that voluntary turnover is also associated with significant linguistic patterns. Notably, the "authentic" linguistic measure shows a significant negative relationship with voluntary turnover. Similarly, the use of the "I pronoun" maintains a strong positive relationship with both types of turnover, emphasizing the potential impact of a self-referential communication style on turnover outcomes.

However, linguistic variables such as "analytic," "they" pronoun, "affective process," "tentative," "power," and "focus present" show weaker or insignificant relationships with voluntary turnover, in contrast to their stronger and more significant effects in predicting forced

² The number of observations differs slightly between Table 3 and Table 4. Table 3 includes more observations because it considers only the last conference call for each CEO and excludes only the final call preceding voluntary turnovers (since the analysis compares forced turnovers to no turnovers). In contrast, Table 4 excludes the last four conference calls prior to voluntary turnovers, ensuring consistency in the sample. This is because Table 4 analyzes linguistic trends over the final four calls leading up to either a forced turnover or no turnover.

turnover. For example, the use of “they” pronoun is significantly negative at the 5% level in Model (1) of Table 3, but it is insignificantly negative in Model (1) of Table 5 ($p\text{-value} = 0.905$). Overall, these results suggest that emotional expression, uncertainty, and a focus on the present in communication may play a more crucial role in scenarios leading to CEO dismissals than in voluntary exits.

In conclusion, our analysis demonstrates that certain CEO linguistic features are significantly associated with forced turnover. The patterns of language used by CEOs evolve as they approach dismissal, with notable shifts in self-referential language, emotional expression, and focus on the present. While these linguistic variables are strongly linked to forced turnover, their relationship with voluntary turnover is weaker, highlighting the distinct communication patterns in each scenario. These findings provide valuable insights into how CEO communication may signal impending leadership changes and contribute to our understanding of the psychological and cognitive dynamics leading to CEO dismissals.

5.2 Propensity score matching and entropy balancing

To ensure that our results are not driven by differences in firm or CEO characteristics, we employ propensity score matching (PSM) and entropy balancing. These methods allow us to create a more comparable control group by matching or reweighting firms without CEO turnover to those with forced turnover, thereby reducing selection bias and improving the validity of our findings.

For the PSM method, we perform a one-to-one match for each fired CEO’s conference call with that of a CEO who remained with the company (no turnover). We select the pairs that are most similar based on the estimated logit model using firm-level characteristics, implementing nearest-neighbor matching with a caliper of 0.01 to minimize the risk of poor matches. The

objective of PSM in our analysis is to compare CEOs' linguistic measures between the treatment group (forced turnover) and the control group (no turnover) while ensuring similar firm performance. We consider firm size, stock return, ROA, and market-to-book ratio, as these are key indicators of firm performance and profitability (e.g., Frye and Pham, 2018; Pukthuanthong et al., 2018; Fan et al., 2021). As a result, 222 out of 224 CEO conference call observations with a forced turnover are successfully matched with conference call observations from CEOs with no turnover. Once treatment and control samples are matched, we examine the verbal measures used during the CEOs' conference calls. The significance of differences in linguistic features between the two groups is assessed using t-statistics.

We present the results of the propensity score matching analysis in Table 6. Panel A reports the outcomes of the initial step, which involves performing a logit regression of the forced CEO turnover variable on firm characteristics such as firm size, stock return, ROA, and market-to-book ratio as noted above. The findings in Panel A show that the treated sample (forced turnover) and the control sample (no turnover) exhibit similar levels of firm characteristics across all three matching specifications. Specification 1 matches firms within the same industry (as defined by the first two digits of the Standard Industrial Classification) and the same fiscal quarter. Specification 2 matches firms within the same industry, while Specification 3 matches firms within the same industry and the same fiscal year. The results are consistent across all three specifications, confirming that the two matched groups (forced turnover and no turnover) have comparable firm performance.

In Panel B of Table 6, we present the differences in linguistic measures between the forced turnover sample and the matched no turnover sample in the column titled "Forced vs. No Turnover." Consistent with the results in Table 3, Panel B shows that CEOs in the forced turnover

group use a higher level of analytical language and a lower level of authenticity (Specification 1 and 3). Their speech also contains a higher frequency of “I” pronoun and a lower frequency of “They” pronoun across all three specifications. Furthermore, fired CEOs use fewer affective words in their conference calls (Specification 2), display lower levels of tentative language (Specifications 1 and 2), and exhibit a greater use of power-related words (Specification 1) and a stronger focus on the present (Specification 3). These findings provide consistent evidence, reinforcing the panel regression results, that dismissed CEOs demonstrate specific linguistic features prior to their dismissal relative to non-departing CEOs.

We further complement the propensity score matching approach by conducting a conditional logit regression on our matched sample. Table 7 presents the results of this regression for the three matched samples. The regressions on that matched samples with similar firm characteristics underscores that these CEOs adopt distinct linguistic styles independent of firm-specific factors, potentially signaling their awareness of an impending dismissal. The findings confirm the presence of CEO-specific verbal communication patterns during conference calls prior to forced turnover. Specifically, we observe that certain linguistic features—such as less frequent use of “They” pronoun, and increased use of present-focused language—are consistently associated with CEOs who experience forced turnover. The negative relationship with authenticity suggests that these CEOs may use a less transparent communication style, possibly to manage perceptions as they navigate challenging circumstances. Additionally, the significant positive association with present-focused language indicates an emphasis on immediate issues, reflecting an acute awareness of current pressures. These patterns align with prior research on linguistic cues discussed in Section 4, illustrating how specific verbal choices can signal an approaching CEO dismissal.

We also conduct an entropy balancing analysis to further validate the robustness of our findings. Table 8 presents the results of this analysis in two panels. Panel A compares performance variables between firms experiencing forced CEO turnover and those without turnover, both before and after weighting. The results show that, after applying entropy balancing, the forced turnover and control samples exhibit similar performance characteristics, ensuring comparability between the two groups. Panel B examines the differences in linguistic measures between CEOs in forced turnover firms and those in no turnover firms after weighting. The results are consistent with those from the propensity score matching approach, further confirming that linguistic features such as lower authenticity, increased use of present-focused language, and reduced use of “They” pronoun are associated with CEOs experiencing forced turnover. This additional analysis supports the robustness of our findings and provides further evidence that linguistic cues can serve as meaningful indicators of an impending CEO dismissal.

In conclusion, both the propensity score matching (PSM) and entropy balancing analyses provide robust support for our findings regarding the relationship between CEO linguistic features and forced turnover. The PSM results demonstrate that the forced turnover and control groups exhibit comparable firm performance, ensuring that observed linguistic differences are not driven by firm-specific characteristics. Similarly, the entropy balancing analysis corroborates these results by aligning performance variables across the two groups and confirming that linguistic features, such as reduced authenticity and increased present-focused language, are consistently associated with CEOs facing forced turnover. Together, these complementary methodologies strengthen the validity of our conclusions, highlighting that distinct verbal communication patterns precede CEO dismissals, independent of firm performance, and providing further evidence of the predictive power of CEO speech in turnover scenarios.

6. Supplementary analysis

The panel regression, propensity score matching, and entropy balancing analyses indicate that CEOs' linguistic patterns during their final conference calls are significantly associated with forced turnover. To further explore these linguistic features, we analyze a sample of newly appointed CEOs following both forced and voluntary turnovers. Using a difference-in-differences regression, we examine changes in verbal communication as firms transition to new CEOs. This analysis reinforces our main findings by demonstrating that CEOs' communication styles shift prior to dismissal while also highlighting how the nature of turnover influences the linguistic patterns of their successors.

Additionally, we investigate CFOs' speech during earnings conference calls, specifically focusing on their language style before a CEO's forced departure. This analysis helps determine whether linguistic shifts are unique to CEOs or if other top executives, such as CFOs, also adjust their speech patterns in anticipation of a leadership change. Finally, as a robustness check, we analyze the question-and-answer (Q&A) portion of earnings calls to compare linguistic measures with those extracted from the management discussion (MD) section. Since CEOs likely have greater control over the content in the MD section, we expect the observed language shifts to be more pronounced there than in the Q&A segment.

Collectively, these supplementary analyses provide further evidence that CEOs modify their language before a forced turnover, suggesting an awareness of their impending dismissal.

6.1 Do new CEOs have distinct communication patterns?

Using a sample of incoming CEOs following forced turnovers, we analyze the linguistic differences between the speech patterns of ousted CEOs during their final conference calls and those of newly appointed CEOs during their first conference call after taking over. From the full sample of CEOs in ExecuComp, we identify a new CEO following each turnover, defining a binary variable equal to one when a new CEO first appears in the firm. This process yields 151 new CEO conference call observations out of 224 forced turnover cases. Table 9 reports differences in language style during conference calls after a new CEO takes over following a turnover. Panel A presents linguistic measures for new CEOs succeeding a forced turnover, while Panel B examines new CEOs following a voluntary turnover.

The findings in Panel A indicate that incoming CEOs are less analytical and exhibit greater authenticity, suggesting a more spontaneous speaking style in their first conference call compared to the final conference call of ousted CEOs. A higher frequency of “I” pronouns and increased use of affective language suggest a shift away from a strategic or less transparent communication style, reinforcing the notion that efforts to manage perceptions may intensify in the final conference calls before a CEO’s dismissal. We also observe that incoming CEOs are more likely to focus on the present than dismissed CEOs. At the same time, we do not observe significant differences in the use of “They” pronoun, tentative language, or power-related words.

Examining new CEOs’ language after voluntary turnovers in Panel B, we find that they exhibit lower tentativeness and use fewer power-related words compared to their retired predecessors. Similar to the final conference calls before a forced turnover, new CEOs’ first conference calls maintain a strong present focus. The findings on “I” pronoun usage and present

focus suggest that new CEOs emphasize immediacy and personal agency, aligning with the need to project control and confidence during their initial public appearances.

Prior research by Bochkay et al. (2019) suggests that newly appointed CEOs tend to exhibit greater optimism early in their tenure, which declines over time. They measure optimism as the difference between the number of positive and negative words, scaled by the total word count. In our analysis, the affective process captures the overall emotional tone in CEOs' speech, and we find that it is significantly higher for incoming CEOs.

To further investigate this heightened emotional expression, – in an unreported table – we examine the positive and negative emotion categories, which are subcomponents of the affective process in LIWC. Our findings indicate that new CEOs use more positive and fewer negative emotion words in their first conference calls following a forced turnover. For instance, the average score for positive emotion words is 4.363 for new CEOs compared to 3.839 for fired CEOs, a difference that is statistically significant at the 1% level. In terms of negative emotion words, new CEOs score 0.438 words versus 0.505 for fired CEOs, a difference that is significant at the 10% level. These patterns suggest that new CEOs tend to communicate with greater optimism and less negativity as they take over leadership.

Next, we examine whether variations in new CEOs' speech patterns are influenced by the type of turnover. To do so, we perform a difference-in-difference analysis to determine whether new CEOs' verbal communication differs significantly following a forced turnover compared to a voluntary turnover. Our sample includes 728 cases of voluntary turnovers and 149 cases of Forced turnovers with observations for new CEOs' conference calls.

The difference-in-difference approach allows us to capture not only changes in linguistic attributes but also the direction of these changes in LIWC-derived measures after a new CEO

assumes control while controlling for firm characteristics. We implement this framework using the full sample of new CEOs following both types of turnovers, comparing linguistic changes between transitions from a fired CEO to a new CEO and from a retired CEO to a new CEO. The following specification outlines our approach:

$$\begin{aligned}
LIWC\ Var_{it} = & \beta_0 + \beta_1 FV_{it} \times Post_{it} + \beta_2 FV_{it} + \beta_3 Post_{it} + \alpha_1 Firm\ Size_{it} + \\
& \alpha_2 ROA_{it} + \alpha_3 MTB_{it} + \alpha_4 Leverage_{it} + \alpha_5 Return_{it} + \alpha_6 Earning\ Surprise_{it} + \alpha_7 Loss_{it} + \\
& \alpha_8 Following_{it} + \delta_1 Time\ FE_t + \delta_2 Firm\ FE_i + \varepsilon_{it} \quad (2)
\end{aligned}$$

In this specification, $LIWC\ Var_{it}$ represents the linguistic measures (“Analytic,” “Authentic,” “I” pronoun, “They” pronoun, “Affective Process,” “Tentative,” “Power,” and “Focus Present”) in the final conference call of departing CEOs and the first conference call of newly appointed CEOs. FV_{it} is a treatment indicator denoting the type of turnover, taking a value of 1 for forced turnovers and 0 for voluntary turnovers. $Post_{it}$ is an indicator variable that equals 1 for new CEOs and 0 for departing CEOs in conference call observations. Other control variables are consistent with those described in Equation (1). The coefficient β_1 represents the difference-in-difference estimator, capturing the impact of forced turnover on the linguistic style of new CEOs relative to those who assume leadership following an ordinary (voluntary) turnover.

The results of the difference-in-difference specification are presented in Table 10 and are generally consistent with the findings in Table 3, which highlight linguistic differences in newly appointed CEOs following a forced turnover. The linguistic features—analytical thinking, “I” pronoun usage, affective process, tentative words, and present focus—retain consistent directional effects. Specifically, newly hired CEOs who take over after a forced turnover exhibit lower analytical thinking, greater use of self-referential words, higher expression of emotion, fewer

tentative words, and an increased focus on the present in their first conference call compared to incoming CEOs following a voluntary turnover.

Overall, these findings suggest that newly appointed CEOs adjust their verbal communication style more noticeably when stepping into a role after a forced turnover than after an orderly transition. Moreover, the results reinforce the earlier evidence that departing CEOs exhibit distinct linguistic patterns before being dismissed. By focusing on linguistic changes after the leadership transition, the difference-in-difference analysis helps isolate the effect of turnover type on new CEOs' communication styles.

5.3 Do CFOs also change their language style?

Earnings conference calls provide an opportunity to analyze the language styles of not only CEOs but also other executives, such as CFOs, who participate in these discussions. To determine whether the linguistic changes observed in CEOs' final conference calls before dismissal are unique to their role, we examine CFOs' speech patterns. Using the matched sample of fired CEOs and non-departing CEOs from firms with similar characteristics (firm size, ROA, stock return, and book-to-market ratio) presented in Table 7, we replace CEOs' speech with that of CFOs.

Out of the 222 matched CEO conference call observations, we identify 213 cases where CFOs participated in the conference calls. By leveraging the matched sample derived from the propensity score matching method, we can analyze differences in CFO communication styles within the same industry and fiscal year. This approach allows us to first explore variations in CFOs' linguistic features between firms that experienced a forced CEO turnover and those that did not. Ultimately, this analysis helps determine whether the observed linguistic changes are specific to CEOs or whether they also extend to other top executives.

Table 11 compares CFO language styles between CFOs in companies with forced CEO turnovers and those in matched companies without turnovers. Each CFO observation from the forced turnover subsample is matched with one from the no-turnover group, with differences reported in the “Forced vs. No” column. The lack of significant results suggests that CFOs do not alter their language style, reinforcing the idea that the observed linguistic shifts are specific to CEOs.³

Next, we investigate whether variations in CFOs’ speech patterns are influenced by the type of turnover. To do this, we analyze their speech patterns using a difference-in-difference approach to examine whether there are significant differences in CFOs’ verbal language following a forced turnover compared to a voluntary turnover. Using the framework in Table 10, we compare changes in CFO linguistic measures between the transitions of fired CEOs to new CEOs and retired CEOs to new CEOs using specification (2). Here, $LIWC\ Var_{it}$ represents linguistic measures (“Analytic,” “Authentic,” “I” pronoun, “They” pronoun, “Affective Process,” “Tentative,” “Power,” and “Focus Present”) in the final conference call for CFOs before and after a CEO turnover.

The results of the difference-in-difference analysis are reported in Table 12. The results from the difference-in-difference analysis, which focuses on linguistic changes after the transition, allow us to better isolate the effects of the type of turnover on CFOs. We again find consistent evidence that CFOs do not change their language style. The findings indicate that CFOs’ verbal communication before a turnover is similar between the first conference calls following both forced and voluntary turnovers. These results align with those presented in Table 11, which reports

³ When we repeat the analysis using matched firms—either from the same industry and fiscal quarter or from the same industry alone—we find no significant change in CFOs’ language style.

the matched sample of CFOs between forced turnover and no turnover. Therefore, our analysis shows that language style changes are specific to CEOs and not CFOs.

6.3 Language styles in Q&A portion of conference calls

We further examine CEO language patterns in the Q&A portion of earnings calls, where responses are less scripted and more reactive. Unlike the management discussion (MD) section, which allows for careful message crafting, the Q&A segment provides a setting where CEOs have less control over their communication, offering a different perspective on linguistic indicators of forced turnover.

Table 13 presents the results of logistic regressions examining the relationship between CEO linguistic measures during the Q&A portion of conference calls and forced CEO turnover. The models include fixed effects for year, and year-quarter, firm, and industry as outlined in Table 3. The results show that while some linguistic measures, such as “analytic” and “I” pronoun, are positively associated with forced turnover, the overall strength of the coefficients is weaker compared to those observed in the management discussion (MD) section, as reported in Table 3. Additionally, several linguistic measures, including “affective process,” “tentative,” “power,” and “focus present,” show no significant association with forced turnover.

These findings suggest that the language used in the MD portion of earnings calls provides a stronger and more consistent signal of potential CEO turnover compared to the Q&A section, where the associations are less reliable. This aligns with the argument that CEOs have more control over the prepared and scripted MD section, whereas the Q&A portion is more reactive and less structured. Therefore, the results highlight that the CEOs’ language style during the MD section

serves as a channel through which they adjust their verbal communication, indicating that they may be aware of their imminent dismissal.

7. Conclusion

We examine how CEO language style changes in the final earnings conference calls before dismissal, revealing linguistic patterns that signal impending forced turnover. Through panel regressions, propensity score matching, entropy balancing, and difference-in-difference analyses, we find that CEOs on the brink of dismissal exhibit distinct speech characteristics, including higher analytical language, increased use of first-person pronouns, and lower authenticity. These patterns persist across multiple analytical approaches, reinforcing the robustness of language as a potential indicator of forced turnover.

Further analyses highlight notable differences in the communication styles of newly appointed CEOs following forced versus voluntary turnovers. After a forced turnover, new CEOs tend to adopt a less analytical and more authentic speech style, marking a clear departure from the strategic, and potentially defensive, language of their predecessors. This shift suggests that the nature of a CEO transition—whether forced or voluntary—has a measurable impact on executive communication.

Additionally, our examination of CFOs' language use suggests that these linguistic shifts are specific to CEOs, as CFOs do not exhibit similar changes before a CEO's dismissal. This distinction highlights the unique pressures CEOs face when managing stakeholder perceptions amid leadership uncertainty. Moreover, our comparison of the management discussion (MD) and Q&A portions of conference calls underscores the role of speech control in shaping communication patterns. While CEOs' language in the MD section shows strong and consistent associations with forced turnover, the Q&A segment—being more reactive and unscripted—exhibits weaker and

less reliable links to dismissal. This finding reinforces the idea that CEOs strategically manage their language when they have greater control over messaging.

By demonstrating that linguistic cues in earnings conference calls can serve as signals of leadership instability, our study contributes to the literature on executive communication and corporate governance. These findings have meaningful implications for investors, analysts, and corporate boards, offering a new dimension for assessing CEO performance and anticipating leadership changes.

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Appendix

Variable	Definition
Forced (Voluntary) Turnover	An indicator variable equal to one if there is a forced (voluntary) turnover between the current and the next conference call, and zero if no turnover occurs during that interval.
Analytic	LIWC measure capturing the degree of formal, logical, and hierarchical thinking in CEOs' speech.
Authentic	LIWC measure assessing the level of honesty, personal disclosure, and genuineness in CEOs' speech.
I Pronoun	LIWC measure indicating the frequency of first-person singular pronoun ("I") usage in CEOs' speech.
They Pronoun	LIWC measure indicating the frequency of third-person plural pronoun ("they") usage in CEOs' speech.
Affective Process	LIWC measure representing the frequency of emotion-related words in CEOs' speech.
Tentative	LIWC measure indicating the degree of uncertainty or hesitation in language (e.g., maybe, perhaps).
Power	LIWC measure capturing the use of words related to dominance, leadership, and influence.
Focus Present	LIWC measure reflecting the emphasis on the present time (e.g., now, today) in speech.
Firm Size	Natural logarithm of total assets at the end of the fiscal quarter.
ROA	Return on assets; earnings before extraordinary items divided by total assets.
MTB	Market-to-book; Market value of equity plus book value of liabilities divided by book value of assets.
Return	Average monthly stock return over the quarter
Leverage	Ratio of long-term debt to total assets.
Loss	Indicator variable equal to 1 if earnings per share (EPS) are negative, and 0 otherwise.
Earnings Surprise	Actual earnings per share minus consensus analyst forecast per share divided by stock price at the end of the quarter.
Following	Natural logarithm of the number of analysts issuing an earnings forecast for the firm.
Institutional Ownership	Percentage of shares held by institutional investors, calculated from 13-F filings.
CEO Tenure	Natural logarithm of the number of years between the CEO's appointment date and the end of the current period.
CEO Age	Natural logarithm of the CEO's age.
CEO Gender	Dummy variable equal to 1 if the CEO is female and 0 if male.
CEO Ownership	Percentage of outstanding shares owned by the CEO.

Table 1. Sample selection

This table provides an overview of our sample construction process and the distribution of CEO turnover events. Panel A reports the total number of firm-year observations and cases of CEO turnover and forced turnovers from 2010 to 2018. Forced CEO turnovers are classified based on Parrino (1997) and Campbell et al. (2011). Panel B summarizes our process of creating the final sample.

Panel A: Distribution of annual CEO turnovers from ExecuComp.

Year	Number of Observations	Number of Turnovers	Number of Forced Turnovers	Forced as % of total Turnovers
2010	2,141	180	38	21.1%
2011	2,118	198	42	21.2%
2012	2,091	192	47	24.5%
2013	2,088	183	30	16.4%
2014	2,074	192	43	22.4%
2015	2,004	198	38	19.2%
2016	1,928	196	38	19.4%
2017	1,876	182	30	16.5%
2018	1,811	185	41	22.2%
Total	18,131	1,706	347	20.3%

Panel B: Sample selection for quarterly CEO turnovers

	Number of Observations	Number of Forced Turnovers	Voluntary Turnovers
Firm-year observations from ExecuComp	18,131	347	1,359
Firm-quarter observations with Compustat and CRSP data	69,087	319	1,303
Firm-quarter observations with conference call transcripts	45,438	224	923

Table 2. Summary statistics

This table reports summary statistics for the linguistic attributes derived from LIWC in the last conference call (Panel A), along with firm-level and CEO-level variables (Panel B), for staying CEOs (No Turnover), forced-out CEOs (Forced Turnover), and the difference between the two groups. Additionally, the table includes the mean and standard deviation of each linguistic measure as reported in the LIWC manual. The grand means represent the unweighted average across six LIWC genres, while the reported standard deviations (Mean SDs) reflect the unweighted average of standard deviations across those genres.

Variables	No Turnover			Forced Turnover			Difference: No turnover and Forced turnover		LIWC Manual	
	Mean (1)	Median (2)	SD (3)	Mean (4)	Median (5)	SD (6)	Mean (7)	p-Value (8)	Grand Mean (9)	Mean SD (10)
<i>Panel A: Linguistic features</i>										
Analytic	91.314	93.110	6.875	91.594	93.225	6.314	-0.279	(0.544)	56.34	17.58
Clout	89.087	90.350	6.537	89.601	90.720	5.903	-0.514	(0.240)	57.95	17.51
Authentic	44.533	43.960	13.485	43.059	41.260	12.526	1.474	(0.103)	49.17	20.92
Tone	81.970	84.940	13.362	81.827	85.795	13.140	0.143	(0.873)	54.22	23.27
Function Words	44.276	43.930	3.826	44.395	43.895	3.537	-0.119	(0.643)	51.87	5.13
I Pronoun	0.613	0.500	0.474	0.700	0.470	0.725	-0.087***	(0.006)	4.99	2.46
We Pronoun	5.833	5.830	1.527	5.912	5.985	1.498	-0.079	(0.441)	0.72	0.83
You Pronoun	0.387	0.300	0.340	0.361	0.250	0.339	0.026	(0.253)	1.70	1.35
She/He Pronoun	0.053	0.000	0.148	0.077	0.000	0.175	-0.025**	(0.013)	1.88	1.53
They Pronoun	0.267	0.190	0.282	0.238	0.170	0.241	0.029	(0.125)	0.66	0.60
Impersonal Pronoun	3.019	2.870	1.106	2.996	2.875	1.111	0.023	(0.754)	5.26	1.62
Prepositions	16.380	16.370	1.302	16.476	16.350	1.188	-0.096	(0.272)	12.93	2.11
Adverbs	2.570	2.370	1.078	2.401	2.225	0.961	0.169**	(0.019)	5.27	1.61
Conjunctions	5.452	5.410	0.991	5.533	5.470	0.956	-0.081	(0.220)	5.90	1.57
Negations	0.249	0.200	0.229	0.232	0.200	0.200	0.017	(0.269)	1.66	0.86
Comparisons	2.750	2.710	0.762	2.786	2.750	0.740	-0.036	(0.478)	2.23	0.95
Interrogatives	0.625	0.580	0.343	0.616	0.575	0.312	0.009	(0.683)	1.61	0.76
Numbers	5.706	5.210	2.693	5.134	4.745	2.384	0.571***	(0.002)	2.12	2.07
Quantifier	2.163	2.110	0.672	2.103	2.060	0.595	0.060	(0.181)	2.02	0.83
Affective Process	4.453	4.370	1.106	4.386	4.400	1.028	0.066	(0.370)	5.57	1.99
Positive Emotion	3.926	3.840	1.078	3.867	3.955	1.001	0.059	(0.411)	3.67	1.63
Negative Emotion	0.503	0.430	0.385	0.500	0.445	0.332	0.003	(0.897)	1.84	1.09
Anxiety	0.094	0.050	0.143	0.090	0.060	0.138	0.004	(0.680)	0.31	0.32
Anger	0.063	0.000	0.109	0.068	0.000	0.120	-0.004	(0.556)	0.54	0.59
Sadness	0.194	0.140	0.219	0.185	0.150	0.173	0.009	(0.538)	0.41	0.40
Social Process	8.947	8.910	2.017	9.085	9.090	1.891	-0.138	(0.307)	9.74	3.38
Cognitive Process	7.076	6.970	1.462	7.075	7.050	1.298	0.002	(0.987)	10.61	3.02
Insight	1.305	1.240	0.580	1.290	1.270	0.518	0.014	(0.717)	2.16	1.08
Discrepancy	0.715	0.660	0.388	0.722	0.690	0.372	-0.007	(0.785)	1.44	0.80
Tentative	1.167	1.070	0.588	1.075	0.990	0.526	0.092**	(0.019)	2.52	1.09
Certainty	1.055	1.010	0.438	1.050	1.060	0.405	0.005	(0.876)	1.35	0.70
Differentiation	1.229	1.150	0.598	1.148	1.055	0.523	0.082*	(0.041)	2.99	1.18
Perceptual Process	1.064	0.980	0.562	1.122	1.070	0.566	-0.059	(0.118)	2.70	1.20

Drives	14.132	14.110	2.441	14.212	14.055	2.334	-0.080	(0.624)	6.93	2.03
Affiliation	6.759	6.730	1.673	6.847	6.805	1.592	-0.088	(0.432)	2.05	1.28
Achieve	3.328	3.270	1.121	3.467	3.500	1.072	-0.139*	(0.064)	1.30	0.82
Power	3.310	3.220	0.969	3.348	3.245	1.000	-0.038	(0.563)	2.35	1.12
Risk	0.324	0.240	0.327	0.322	0.235	0.337	0.002	(0.927)	0.47	0.41
Focus Past	2.188	2.110	0.803	2.106	1.980	0.692	0.082	(0.126)	4.64	2.06
Focus Present	6.462	6.300	1.690	6.598	6.525	1.598	-0.136	(0.229)	9.96	2.80
Focus Future	1.612	1.560	0.640	1.711	1.590	0.698	-0.099**	(0.021)	1.42	0.90
Informal	0.288	0.240	0.237	0.275	0.215	0.303	0.013	(0.418)	2.52	1.65

Panel B: Firm and CEO characteristics

Variables	No Turnover			Forced Turnover			Difference: No turnover and Forced turnover	
	Mean	Median	SD	Mean	Median	SD	Mean	p-Value
Firm Size	7.964	7.879	1.744	7.741	7.525	1.914	0.223*	(0.057)
ROA	0.011	0.011	0.025	0.002	0.006	0.031	0.008***	(0.000)
Earning Surprise	0.001	0.000	0.037	0.000	0.000	0.015	0.000	(0.849)
Loss	0.083	0.000	0.276	0.129	0.000	0.336	-0.047**	(0.012)
MTB	2.004	1.551	1.342	1.761	1.400	1.218	0.243***	(0.007)
Leverage	0.224	0.197	0.195	0.245	0.223	0.207	-0.021	(0.122)
Analyst Following	2.233	2.197	0.667	2.205	2.197	0.695	0.028	(0.552)
Return	0.013	0.013	0.060	-0.009	-0.005	0.079	0.022***	(0.000)
Tenure	1.862	1.909	0.817	1.736	1.719	0.582	0.126**	(0.021)
Ownership	0.021	0.006	0.061	0.008	0.004	0.011	0.013***	(0.001)
Age	4.043	4.043	0.119	3.991	4.007	0.095	0.052***	(0.000)
Gender	0.047	0.000	0.211	0.068	0.000	0.252	-0.021	(0.138)
Institutional Ownership	0.813	0.854	0.171	0.805	0.848	0.196	0.008	(0.523)

Table 3. Linguistic variables and forced CEO turnover

This table presents estimated coefficients from a logistic regression of LIWC-derived linguistic measures in CEOs' earnings conference calls on the likelihood of a forced CEO turnover, using the Management Discussion (MD) section of the calls. The four models incorporate various fixed effects, including firm, fiscal year, fiscal quarter, year-quarter, and industry fixed effects. Reported statistics are based on robust standard errors clustered at the firm level. P-values are shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Forced CEO Turnover			
	Model (1)	Model (2)	Model (3)	Model (4)
<i>LIWC linguistic variables</i>				
Analytic	0.103*** (0.000)	0.043** (0.039)	0.043** (0.045)	0.042** (0.045)
Authentic	-0.023** (0.018)	-0.017*** (0.007)	-0.016** (0.013)	-0.017*** (0.010)
I Pronoun	0.932*** (0.000)	0.399** (0.012)	0.379** (0.022)	0.397** (0.014)
They Pronoun	-1.067** (0.035)	-0.721** (0.022)	-0.715** (0.025)	-0.710** (0.021)
Affective Process	-0.275** (0.030)	-0.154* (0.063)	-0.144* (0.084)	-0.140* (0.091)
Tentative	-0.430* (0.069)	-0.220 (0.184)	-0.212 (0.200)	-0.268 (0.111)
Power	0.621*** (0.000)	0.214** (0.035)	0.213** (0.038)	0.213** (0.035)
Focus Present	0.300*** (0.004)	0.195** (0.010)	0.198** (0.011)	0.195** (0.011)
<i>Control variables</i>				
Firm Size	-0.013 (0.975)	-0.143 (0.141)	-0.149 (0.128)	-0.125 (0.188)
ROA	-7.824** (0.050)	-9.384*** (0.004)	-9.176*** (0.005)	-9.226*** (0.003)
Earnings Surprise	0.349 (0.905)	0.425 (0.557)	0.497 (0.543)	0.504 (0.499)
Loss	-0.607 (0.157)	-0.085 (0.777)	-0.058 (0.848)	-0.084 (0.781)
MTB	-0.397** (0.028)	-0.179 (0.103)	-0.182* (0.099)	-0.146 (0.152)
Leverage	2.337* (0.057)	1.003* (0.072)	1.010* (0.068)	1.085** (0.047)
Following	0.335 (0.414)	0.220 (0.281)	0.222 (0.278)	0.188 (0.342)
Return	-1.344 (0.399)	-3.688** (0.012)	-3.746** (0.013)	-3.549** (0.012)
Tenure	2.592*** (0.000)	0.407*** (0.000)	0.395*** (0.000)	0.388*** (0.000)

Ownership	-90.511*** (0.000)	-35.875*** (0.001)	-35.867*** (0.001)	-36.267*** (0.001)
Age	-5.685*** (0.000)	-4.293*** (0.000)	-4.205*** (0.000)	-3.893*** (0.000)
Gender	0.758 (0.223)	0.678** (0.036)	0.671** (0.038)	0.650** (0.044)
Institutional Ownership	-0.241 (0.843)	-0.571 (0.306)	-0.561 (0.312)	-0.383 (0.484)
Constant		7.763** (0.041)	6.812* (0.086)	6.359* (0.099)
Industry FEs		Yes	Yes	Yes
Year FEs		Yes		
Quarter FEs				Yes
Year-Quarter FEs	Yes		Yes	
Firm FEs	Yes			
Observations	3,859	33,138	33,136	33,138
Pseudo R ²	0.278	0.100	0.113	0.081

Table 4. Trend analyses of linguistic variables and forced CEO turnover

This table presents a trend analysis of estimated coefficients from logistic regressions of LIWC-derived linguistic measures in CEOs' earnings conference calls on the likelihood of a forced CEO turnover. The analysis uses the Management Discussion (MD) section of calls from the fourth conference call before dismissal (-4) through to the last conference call before dismissal (-1). The four models include fixed effects for firm, fiscal year, fiscal quarter, year-quarter, and industry. Reported statistics are based on robust standard errors clustered at the firm level. P-values are shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Forced CEO turnover using conference call (-4)

Variables	Model (1)	Model (2)	Model (3)	Model (4)
Analytic	-0.017 (0.521)	-0.026 (0.221)	-0.021 (0.346)	-0.021 (0.324)
Authentic	-0.008 (0.421)	-0.005 (0.533)	-0.005 (0.520)	-0.005 (0.473)
I Pronoun	-0.301 (0.373)	-0.380 (0.161)	-0.392 (0.154)	-0.375 (0.168)
They Pronoun	-0.706 (0.126)	-0.232 (0.499)	-0.246 (0.473)	-0.230 (0.498)
Affective Process	-0.148 (0.278)	-0.053 (0.564)	-0.055 (0.549)	-0.063 (0.500)
Tentative	0.143 (0.573)	-0.039 (0.821)	-0.025 (0.884)	-0.022 (0.894)
Power	-0.126 (0.412)	-0.180 (0.142)	-0.177 (0.147)	-0.172 (0.165)
Focus Present	0.009 (0.935)	-0.018 (0.815)	0.002 (0.983)	0.004 (0.960)
Controls	Yes	Yes	Yes	Yes
Industry FEs		Yes	Yes	Yes
Year FEs		Yes		
Quarter FEs				Yes
Year-Quarter FEs	Yes		Yes	
Firm FEs	Yes			
Observations	3,018	30,436	29,764	30,782
Pseudo R ²	0.228	0.062	0.087	0.064

Panel B: Forced CEO turnover using conference call (-3)

Variables	Model (1)	Model (2)	Model (3)	Model (4)
Analytic	0.053 (0.106)	0.011 (0.652)	0.009 (0.706)	0.008 (0.730)
Authentic	-0.011 (0.299)	-0.003 (0.639)	-0.005 (0.494)	-0.005 (0.514)
I Pronoun	-0.522 (0.177)	-0.501** (0.044)	-0.503** (0.041)	-0.506** (0.042)
They Pronoun	-0.579	-0.211	-0.240	-0.245

	(0.226)	(0.557)	(0.498)	(0.490)
Affective Process	-0.347**	-0.109	-0.113	-0.119
	(0.014)	(0.221)	(0.202)	(0.180)
Tentative	-0.365	-0.348**	-0.360**	-0.368**
	(0.161)	(0.040)	(0.032)	(0.025)
Power	0.175	-0.111	-0.109	-0.106
	(0.265)	(0.269)	(0.283)	(0.294)
Focus Present	0.178	0.099	0.093	0.090
	(0.121)	(0.182)	(0.212)	(0.227)
Controls	Yes	Yes	Yes	Yes
Industry FEs		Yes	Yes	Yes
Year FEs		Yes		
Quarter FEs				Yes
Year-Quarter FEs	Yes		Yes	
Firm FEs	Yes			
Observations	3,061	30,851	30,056	30,851
Pseudo R ²	0.275	0.072	0.091	0.072

Panel C: Forced CEO turnover using conference call (-2)

Variables	Model (1)	Model (2)	Model (3)	Model (4)
Analytic	0.089***	0.036*	0.031	0.031
	(0.010)	(0.066)	(0.115)	(0.112)
Authentic	-0.009	-0.003	-0.002	-0.002
	(0.409)	(0.668)	(0.739)	(0.734)
I Pronoun	0.206	-0.112	-0.119	-0.120
	(0.500)	(0.675)	(0.655)	(0.654)
They Pronoun	-0.639	-0.193	-0.199	-0.203
	(0.197)	(0.556)	(0.543)	(0.530)
Affective Process	-0.542***	-0.171*	-0.171*	-0.169*
	(0.000)	(0.073)	(0.071)	(0.075)
Tentative	-0.367	-0.228	-0.247	-0.266*
	(0.154)	(0.151)	(0.120)	(0.090)
Power	0.230	-0.064	-0.074	-0.077
	(0.132)	(0.512)	(0.455)	(0.435)
Focus Present	0.226**	0.170**	0.151**	0.150**
	(0.047)	(0.014)	(0.033)	(0.031)
Controls	Yes	Yes	Yes	Yes
Industry FEs		Yes	Yes	Yes
Year FEs		Yes		
Quarter FEs				Yes
Year-Quarter FEs	Yes		Yes	
Firm FEs	Yes			
Observations	3,130	30,859	30,131	30,859
Pseudo R ²	0.286	0.0725	0.085	0.072

Panel D: Forced CEO turnover using conference call (-1)

Variables	Model (1)	Model (2)	Model (3)	Model (4)
Analytic	0.121*** (0.000)	0.043** (0.041)	0.043** (0.048)	0.042* (0.051)
Authentic	-0.024** (0.017)	-0.018*** (0.007)	-0.017** (0.011)	-0.017*** (0.01)
I Pronoun	1.018*** (0.000)	0.395** (0.014)	0.376** (0.025)	0.391** (0.016)
They Pronoun	-1.253** (0.025)	-0.751** (0.02)	-0.749** (0.021)	-0.750** (0.017)
Affective Process	-0.392*** (0.006)	-0.151* (0.075)	-0.141* (0.098)	-0.138 (0.104)
Tentative	-0.496* (0.062)	-0.227 (0.179)	-0.219 (0.190)	-0.276 (0.105)
Power	0.696*** (0.000)	0.217** (0.034)	0.214** (0.039)	0.218** (0.033)
Focus Present	0.362*** (0.001)	0.200*** (0.01)	0.201** (0.011)	0.199** (0.011)
Controls	Yes	Yes	Yes	Yes
Industry FEs		Yes	Yes	Yes
Year FEs		Yes		
Quarter FEs				Yes
Year-Quarter FEs	Yes		Yes	
Firm FEs	Yes			
Observations	3,229	30,862	30,860	30,862
Pseudo R ²	0.367	0.102	0.115	0.083

Table 5. Linguistic variables and voluntary CEO turnover

This table presents the estimated coefficients from logistic regressions of LIWC-derived linguistic measures in CEOs' earnings conference calls on the likelihood of a voluntary CEO turnover, using the Management Discussion (MD) section of the calls. The four models include fixed effects for firm, fiscal year, fiscal quarter, year-quarter, and industry. Reported statistics are based on robust standard errors clustered at the firm level. P-values are shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Model (1)	Model (2)	Model (3)	Model (4)
<i>LIWC linguistic variables</i>				
Analytic	-0.000 (0.987)	0.028*** (0.007)	0.027*** (0.009)	0.027*** (0.009)
Authentic	-0.021*** (0.000)	-0.011*** (0.003)	-0.010*** (0.006)	-0.010*** (0.004)
I Pronoun	1.204*** (0.000)	1.082*** (0.000)	1.085*** (0.000)	1.082*** (0.000)
They Pronoun	-0.030 (0.905)	-0.114 (0.509)	-0.079 (0.646)	-0.089 (0.602)
Affective Process	0.066 (0.350)	0.077* (0.081)	0.082* (0.065)	0.088** (0.047)
Tentative	-0.211 (0.124)	-0.239*** (0.004)	-0.237*** (0.004)	-0.237*** (0.004)
Power	0.301*** (0.000)	0.270*** (0.000)	0.265*** (0.000)	0.267*** (0.000)
Focus Present	0.051 (0.388)	0.046 (0.236)	0.041 (0.298)	0.042 (0.281)
<i>Control variables</i>				
Firm Size	-0.263 (0.403)	0.069 (0.173)	0.068 (0.180)	0.072 (0.153)
ROA	-4.326 (0.183)	-3.367 (0.157)	-3.503 (0.135)	-3.358 (0.148)
Earnings Surprise	-1.136 (0.667)	-0.481 (0.118)	-0.376 (0.311)	-0.404 (0.281)
Loss	-0.242 (0.434)	-0.189 (0.361)	-0.178 (0.401)	-0.175 (0.398)
MTB	-0.828*** (0.000)	-0.113*** (0.041)	-0.109** (0.045)	-0.106** (0.048)
Leverage	0.623 (0.511)	0.366 (0.254)	0.359 (0.264)	0.373 (0.242)
Following	0.200 (0.506)	-0.075 (0.509)	-0.084 (0.462)	-0.090 (0.420)
Return	0.598 (0.560)	-0.776 (0.320)	-0.460 (0.582)	-0.747 (0.336)
Tenure	5.283*** (0.000)	0.685*** (0.000)	0.678*** (0.000)	0.680*** (0.000)
Ownership	-41.572***	-16.091***	-15.978***	-16.192***

	(0.000)	(0.001)	(0.001)	(0.001)
Age	26.264***	6.986***	7.017***	7.056***
	(0.000)	(0.000)	(0.000)	(0.000)
Gender	0.218	0.158	0.147	0.145
	(0.721)	(0.503)	(0.535)	(0.538)
Institutional Ownership	0.196	0.284	0.272	0.346
	(0.844)	(0.430)	(0.454)	(0.328)
Constant		-37.390***	-38.630***	-37.722***
		(0.000)	(0.000)	(0.000)
Industry Effects		Yes	Yes	Yes
Year Effects		Yes		
Quarter Effects				Yes
Year-Quarter Effect	Yes		Yes	
Firm Effects	Yes			
Observations	15,620	33,157	33,157	33,157
Pseudo R ²	0.594	0.180	0.193	0.185

Table 6. Propensity score matching

This table identifies a control sample of firms with non-departing CEOs using a propensity score matching procedure. The propensity score is estimated within industry, fiscal year, and fiscal quarter using three different models. We use firm characteristics—firm size, ROA, return, and MTB—as covariates, and Panel A presents the firm performance variables for the matched samples. Panel B compares the language styles of the fired CEO subsample and the non-departing CEO subsample. Each dismissed CEO observation is matched with one non-departing CEO observation using nearest neighbor matching with a caliper of 0.01. We then compare the means of LIWC linguistic measures between the two CEO groups (No Turnover – Forced Turnover) and report the differences in the column titled “Forced vs. No.” *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Comparison of covariates in balanced samples

Variables	Matched samples (industry and fiscal quarter)				Matched samples (industry)				Matched samples (industry and fiscal year)			
	Forced Turnover	No Turnover	Forced vs. No	p-value	Forced Turnover	No Turnover	Forced vs. No	p-value	Forced Turnover	No Turnover	Forced vs. No	p-value
Firm Size	7.749	7.614	0.135	(0.444)	7.749	7.672	0.077	(0.667)	7.749	7.765	-0.016	(0.932)
Return	-0.007	-0.008	0.001	(0.806)	-0.007	-0.013	0.006	(0.302)	-0.007	-0.010	0.003	(0.599)
ROA	0.003	0.006	-0.003	(0.255)	0.003	0.007	-0.004*	(0.094)	0.003	0.003	0.000	(0.972)
MTB	1.764	1.757	0.007	(0.944)	1.764	1.812	-0.048	(0.651)	1.764	1.851	-0.087	(0.447)
Observations	222	222			222	222			222	222		

Panel B: LIWC measures of forced CEO turnover and no turnover

Variables	Matched samples (industry and fiscal quarter)				Matched samples (industry)				Matched samples (industry and fiscal year)			
	Forced Turnover	No Turnover	Forced vs. No	p-value	Forced Turnover	No Turnover	Forced vs. No	p-value	Forced Turnover	No Turnover	Forced vs. No	p-value
Analytic	91.586	90.371	1.215*	(0.090)	91.586	91.315	0.271	(0.680)	91.586	91.617	-0.031	(0.956)
Authentic	43.103	44.411	-1.308	(0.284)	43.103	45.097	-1.994	(0.114)	43.103	45.534	-2.431*	(0.055)
I Pronoun	0.692	0.661	0.031	(0.616)	0.692	0.596	0.095*	(0.095)	0.692	0.647	0.045	(0.432)
They Pronoun	0.235	0.298	-0.062***	(0.009)	0.235	0.327	-0.092***	(0.001)	0.235	0.288	-0.052*	(0.058)
Affective Process	4.378	4.323	0.055	(0.591)	4.378	4.552	-0.174*	(0.090)	4.378	4.382	-0.005	(0.961)
Tentative	1.077	1.167	-0.090*	(0.073)	1.077	1.183	-0.106*	(0.062)	1.077	1.124	-0.047	(0.358)
Power	3.341	3.116	0.225**	(0.014)	3.341	3.256	0.085	(0.350)	3.341	3.371	-0.030	(0.765)
Focus Present	6.606	6.635	-0.029	(0.858)	6.606	6.368	0.237	(0.115)	6.606	6.288	0.319**	(0.034)

Table 7. Conditional logistic regression results

This table presents the results of conditional logit regressions conducted on three matched samples of CEOs, where matching was based on firm characteristics. The dependent variable in these regressions is an indicator for whether the CEO experienced a forced turnover. The analysis investigates whether specific linguistic features are associated with a higher likelihood of forced CEO dismissal, even after controlling for firm-level factors. P-values are shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Matched sample (industry and fiscal quarter)	Matched sample (industry)	Matched sample (industry and fiscal year)
<i>LIWC linguistic variables</i>			
Analytic	0.063 (0.118)	0.111** (0.032)	0.113** (0.049)
Authentic	-0.010 (0.412)	-0.032* (0.052)	0.012 (0.423)
I Pronoun	0.255 (0.453)	0.676 (0.238)	-0.476 (0.435)
They Pronoun	-1.880*** (0.009)	-4.472*** (0.000)	-1.586** (0.044)
Affective Process	0.048 (0.781)	-0.742*** (0.004)	0.134 (0.543)
Tentative	-0.817** (0.026)	-1.131** (0.021)	0.026 (0.945)
Power	-0.010 (0.964)	0.370* (0.092)	-0.117 (0.482)
Focus Present	0.213 (0.125)	0.768*** (0.000)	0.564*** (0.003)
<i>Control variables</i>			
Firm Size	-3.348** (0.010)	0.161 (0.454)	0.032 (0.917)
ROA	-368.908*** (0.008)	-2.314 (0.824)	11.436 (0.656)
Earnings Surprise	-13.248 (0.422)	7.072 (0.556)	20.741 (0.185)
Loss	0.242 (0.704)	0.191 (0.772)	0.517 (0.474)
MTB	-5.998** (0.011)	-0.067 (0.746)	0.021 (0.971)
Leverage	1.700 (0.109)	0.855 (0.392)	0.851 (0.412)
Following	0.126 (0.736)	0.053 (0.924)	0.352 (0.341)
Return	-200.360** (0.012)	10.935*** (0.009)	11.574 (0.401)
Tenure	0.829***	0.992**	1.295***

	(0.005)	(0.011)	(0.006)
Ownership	-41.606**	-18.242*	-37.334***
	(0.018)	(0.056)	(0.008)
Age	-4.911***	-5.514**	-11.070***
	(0.006)	(0.011)	(0.001)
Gender	0.493	1.515*	3.689**
	(0.447)	(0.068)	(0.015)
Institutional Ownership	1.196	-1.852	0.697
	(0.315)	(0.192)	(0.580)
Observations	278	236	250
Pseudo R ²	0.293	0.411	0.369

Table 8: Entropy balancing

This table presents the covariate balance before and after applying entropy balancing. Panel A compares performance variables between firms that experienced a forced CEO turnover and those that did not, both before and after weighting. Panel B examines the differences in linguistic measures between CEOs of forced turnover firms and those of no-turnover firms, after weighting. The four models include fixed effects for firm, fiscal year, fiscal quarter, year-quarter, and industry. Reported statistics are based on robust standard errors clustered at the firm level. P-values are shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Comparison of covariates between forced turnover and no turnover: before and after weighting

Variables	Forced Turnover	No Turnover (Unweighted)	No Turnover (Weighted)
Firm Size	7.741	7.964	7.741
Return	0.002	0.011	0.002
ROA	1.761	2.004	1.761
MTB	-0.009	0.013	-0.009
Observations	224	44,204	44,112

Panel B: LIWC measures of forced CEO turnover and no turnover in the weighted sample

Variables	Model (1)	Model (2)	Model (3)
<i>LIWC linguistic variables</i>			
Analytic	0.050** (0.024)	0.051** (0.022)	0.044** (0.045)
Authentic	-0.013* (0.056)	-0.015** (0.043)	-0.013* (0.059)
I Pronoun	0.504*** (0.007)	0.625*** (0.001)	0.444** (0.016)
They Pronoun	-0.931** (0.011)	-0.852** (0.023)	-0.890** (0.013)
Affective Process	-0.193** (0.028)	-0.219** (0.011)	-0.126 (0.165)
Tentative	-0.302* (0.060)	-0.354** (0.021)	-0.337** (0.043)
Power	0.161* (0.097)	0.189* (0.050)	0.120 (0.246)
Focus Present	0.199** (0.012)	0.188** (0.014)	0.157** (0.046)
<i>Control variables</i>			
Firm Size	-0.058 (0.511)	-0.078 (0.372)	-0.028 (0.749)
ROA	-4.233 (0.211)	-2.833 (0.377)	-3.591 (0.274)
Earnings Surprise	0.873 (0.446)	0.695 (0.226)	0.731 (0.365)
Loss	-0.334 (0.271)	-0.319 (0.275)	-0.292 (0.346)

MTB	-0.049 (0.500)	-0.080 (0.248)	-0.059 (0.407)
Leverage	1.417*** (0.005)	1.451*** (0.005)	1.496*** (0.002)
Following	0.154 (0.422)	0.125 (0.495)	0.201 (0.280)
Return	0.163 (0.908)	0.191 (0.898)	0.719 (0.581)
Tenure	0.394*** (0.002)	0.412*** (0.001)	0.380*** (0.002)
Ownership	-32.362*** (0.000)	-36.215*** (0.000)	-33.637*** (0.000)
Age	-5.497*** (0.000)	-5.368*** (0.000)	-5.238*** (0.000)
Gender	0.734** (0.030)	0.755** (0.030)	0.672** (0.040)
Institutional Ownership	-0.234 (0.680)	-0.421 (0.447)	-0.237 (0.666)
Constant	16.095*** (0.000)	15.605*** (0.000)	16.010*** (0.000)
Industry FEs	Yes	Yes	Yes
Year FEs	Yes		
Quarter FEs			Yes
Year-Quarter FEs		Yes	
Observations	33,138	33,136	33,138
Pseudo R ²	0.183	0.217	0.157

Table 9: New CEO analysis

This table reports differences in linguistic measures for incoming CEOs following turnover events. For each type of turnover, we identify the new CEOs and compare the mean LIWC linguistic measures between departing and incoming CEOs. Panel A presents the results for forced turnovers, comparing fired CEOs to their successors (149 observations), with differences reported in the column titled “New CEO vs. Fired CEO.” Panel B presents the results for voluntary turnovers, comparing retired CEOs to their successors (728 observations), with differences reported in the column titled “New CEO vs. Retired CEO.” *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Comparison of LIWC linguistic measures between Fired CEOs and New CEOs

<i>LIWC linguistic variables</i>	New CEO	Fired CEO	New CEO vs. Fired CEO	p-value
Analytic	88.851	91.132	- 2.281***	(0.003)
Authentic	45.884	44.149	1.735	(0.217)
I Pronoun	1.363	0.713	0.650***	(0.000)
They Pronoun	0.282	0.247	0.035	(0.240)
Affective Process	4.557	4.314	0.242**	(0.039)
Tentative	1.141	1.125	0.016	(0.797)
Power	3.211	3.350	-0.140	(0.151)
Focus Present	7.505	6.699	0.806***	(0.000)

Panel B: Comparison of LIWC linguistic measures between Retired CEOs and New CEOs

<i>LIWC linguistic variables</i>	New CEO	Retired CEO	New CEO vs. Retired CEO	p-value
Analytic	91.064	90.202	0.862**	(0.020)
Authentic	45.513	44.980	0.547	(0.399)
I Pronoun	0.889	0.968	-0.078*	(0.072)
They Pronoun	0.256	0.250	0.005	(0.709)
Affective Process	4.655	4.683	-0.029	(0.644)
Tentative	1.068	1.192	-0.124***	(0.000)
Power	3.385	3.592	-0.207***	(0.000)
Focus Present	6.698	6.696	0.002	(0.981)

Table 10: Difference-in-difference analysis of new CEOs

This table presents the difference-in-differences (DID) regression results comparing new CEOs following forced versus voluntary turnovers. In each column, the dependent variable is one of the eight LIWC linguistic variables measured in the final conference call of the departing CEO and the first conference call of the new CEO after the previous CEO's departure. *FV* is an indicator variable equal to one for forced turnovers and zero for voluntary turnovers. *Post* is an indicator variable equal to one for the first conference call of the incoming CEO and zero otherwise. Reported statistics are based on robust standard errors clustered at the firm level, with p-values shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Analytic	Authentic	I Pronoun	They Pronoun	Affective process	Tentative	Power	Focus Present
FV	1.409 (0.280)	-2.655 (0.292)	-0.333* (0.064)	0.005 (0.925)	-0.362* (0.092)	-0.089 (0.406)	-0.103 (0.578)	-0.167 (0.617)
Post	1.198** (0.024)	0.894 (0.348)	-0.107 (0.153)	-0.012 (0.551)	-0.052 (0.546)	-0.100** (0.030)	-0.244*** (0.002)	-0.094 (0.462)
FV×Post	-4.122*** (0.000)	1.774 (0.396)	0.829*** (0.000)	0.018 (0.674)	0.316* (0.053)	0.156* (0.069)	0.084 (0.529)	0.962*** (0.000)
<i>Control variables</i>								
Firm Size	1.401 (0.576)	1.687 (0.667)	-0.214 (0.426)	-0.112 (0.227)	-0.049 (0.888)	-0.270 (0.283)	-0.060 (0.817)	-0.891 (0.108)
ROA	21.201 (0.182)	26.931 (0.428)	-1.540 (0.506)	-0.163 (0.762)	-0.109 (0.964)	-0.643 (0.681)	2.820 (0.189)	-5.549 (0.179)
Earnings Surprise	-2.082 (0.817)	-1.415 (0.921)	1.484 (0.153)	0.169 (0.557)	0.111 (0.923)	-0.340 (0.667)	-0.950 (0.426)	1.019 (0.614)
Loss	0.984 (0.438)	2.331 (0.297)	0.218 (0.199)	-0.006 (0.898)	0.029 (0.881)	-0.025 (0.835)	0.002 (0.992)	-0.237 (0.436)
MTB	0.179 (0.760)	0.498 (0.715)	-0.098 (0.381)	0.016 (0.568)	-0.033 (0.780)	-0.007 (0.910)	-0.055 (0.560)	-0.185 (0.388)
Leverage	1.082 (0.853)	-1.409 (0.875)	-0.184 (0.792)	0.115 (0.592)	0.658 (0.394)	0.082 (0.890)	0.040 (0.962)	0.995 (0.432)
Following	1.883 (0.178)	3.997* (0.082)	-0.075 (0.656)	-0.057 (0.311)	-0.017 (0.944)	0.029 (0.829)	-0.006 (0.974)	-0.312 (0.372)
Return	3.087 (0.515)	-5.633 (0.497)	-0.245 (0.680)	-0.005 (0.976)	0.261 (0.733)	-0.722* (0.074)	-0.412 (0.471)	-0.920 (0.447)

Constant	73.526*** (0.000)	23.715 (0.456)	3.007 (0.162)	1.290* (0.083)	4.979* (0.083)	3.430* (0.082)	3.995** (0.044)	14.665*** (0.001)
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,547	1,547	1,547	1,547	1,547	1,547	1,547	1,547
R ²	0.612	0.645	0.539	0.621	0.653	0.610	0.633	0.636

Table 11: CFO Analysis

In this table, we identify a control sample of firms with non-departing CEOs using a propensity score matching procedure, as discussed in Table 9, to compare the language styles of CFOs in firms with fired CEOs to those in matched firms (within industry and fiscal year) with non-departing CEOs. Each CFO observation in the forced turnover subsample is matched with one CFO observation in the no-turnover subsample using nearest neighbor matching with a caliper of 0.01, which produces 213 matched observations.

LIWC linguistic variables	Forced Turnover	No Turnover	Forced vs. No	p-value
Analytic	96.920	96.926	0.006	(0.986)
Authentic	40.620	41.966	1.347	(0.371)
I Pronoun	0.406	0.355	-0.052	(0.322)
They Pronoun	0.056	0.060	0.004	(0.653)
Affective Process	3.553	3.698	0.145	(0.135)
Tentative	1.316	1.352	0.036	(0.541)
Power	3.030	3.035	0.006	(0.954)
Focus Present	4.517	4.376	-0.141	(0.986)

Table 12: Difference-in-difference analysis of CFOs

This table reports the difference-in-differences (DID) regression results for the CFO sample, comparing language use following forced versus voluntary CEO turnovers. In each column, the dependent variable is one of eight LIWC linguistic variables measured in the final conference call for a CFO under a departing CEO and the first conference call for the CFO after the CEO's departure. *FV* is an indicator variable equal to one for forced turnovers and zero for voluntary turnovers. *Post* is an indicator variable equal to one for the first conference call following the CEO departure and zero otherwise. Reported statistics are based on robust standard errors clustered at the firm level, with p-values shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	Analytic	Authentic	I Pronoun	They Pronoun	Affective process	Tentative	Power	Focus Present
FV	-0.259 (0.493)	0.081 (0.970)	0.020 (0.663)	-0.006 (0.721)	0.084 (0.616)	0.018 (0.843)	0.166 (0.266)	0.107 (0.617)
Post	0.062 (0.766)	0.169 (0.817)	-0.002 (0.933)	-0.008 (0.232)	0.002 (0.975)	-0.020 (0.501)	0.011 (0.793)	-0.112 (0.103)
FV×Post	0.277 (0.647)	0.482 (0.774)	0.026 (0.588)	0.018 (0.308)	0.003 (0.982)	-0.011 (0.885)	0.051 (0.643)	0.162 (0.344)
<i>Control variables</i>								
Firm Size	-1.302 (0.306)	1.020 (0.780)	0.162 (0.149)	-0.005 (0.833)	-0.263 (0.296)	0.014 (0.927)	-0.112 (0.653)	0.381 (0.334)
ROA	5.315 (0.667)	38.945 (0.192)	-0.146 (0.850)	0.097 (0.757)	-0.842 (0.694)	-0.301 (0.782)	1.167 (0.480)	-1.647 (0.647)
Earnings Surprise	7.783 (0.461)	-32.449* (0.092)	-0.523 (0.357)	-0.133 (0.568)	1.969 (0.215)	0.320 (0.598)	0.347 (0.780)	-1.018 (0.710)
Loss	0.167 (0.699)	-0.100 (0.967)	-0.085* (0.074)	0.011 (0.590)	0.165 (0.316)	0.061 (0.561)	-0.067 (0.513)	-0.180 (0.457)
MTB	-0.374 (0.334)	-1.575 (0.201)	0.043 (0.232)	0.005 (0.657)	0.006 (0.950)	-0.058 (0.345)	0.049 (0.638)	0.049 (0.742)
Leverage	0.692 (0.768)	-4.652 (0.609)	-0.462** (0.034)	0.039 (0.578)	-0.763 (0.263)	-0.612 (0.141)	-0.468 (0.418)	-0.075 (0.953)
Following	-0.249 (0.655)	-0.310 (0.898)	-0.090 (0.210)	0.001 (0.976)	0.184 (0.461)	0.028 (0.808)	-0.087 (0.656)	-0.349 (0.235)
Return	-0.624 (0.778)	3.005 (0.701)	0.129 (0.475)	0.004 (0.964)	-0.171 (0.788)	0.078 (0.835)	-0.282 (0.558)	-0.180 (0.826)

Constant	108.205*** (0.000)	35.536 (0.231)	-0.656 (0.465)	0.091 (0.666)	5.576** (0.011)	1.481 (0.211)	4.039** (0.042)	2.196 (0.486)
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,703	1,703	1,703	1,703	1,703	1,703	1,703	1,703
R ²	0.685	0.786	0.640	0.652	0.758	0.778	0.825	0.781

Table 13. Logistic regression results for question-and-answer (Q&A) part

This table presents estimated coefficients from a logistic regression of LIWC-derived linguistic measures in CEOs' earnings conference calls on the likelihood of a forced CEO turnover, using the question-and-answer (Q&A) section of the calls. The four models incorporate various fixed effects, including firm, fiscal year, fiscal quarter, year-quarter, and industry fixed effects. Reported statistics are based on robust standard errors clustered at the firm level. P-values are shown in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	Forced CEO Turnover			
	Model (1)	Model (2)	Model (3)	Model (4)
<i>LIWC linguistic variables</i>				
Analytic	0.009 (0.444)	0.017* (0.071)	0.018* (0.062)	0.015 (0.104)
Authentic	-0.005 (0.582)	-0.016** (0.023)	-0.016** (0.026)	-0.016** (0.032)
I Pronoun	0.226 (0.144)	0.221* (0.071)	0.218* (0.072)	0.205* (0.091)
They Pronoun	-0.426* (0.098)	-0.505*** (0.009)	-0.492** (0.010)	-0.517*** (0.008)
Affective Process	0.125 (0.168)	-0.042 (0.598)	-0.038 (0.631)	-0.039 (0.622)
Tentative	0.083 (0.503)	-0.057 (0.616)	-0.057 (0.615)	-0.063 (0.574)
Power	-0.133 (0.396)	-0.092 (0.488)	-0.090 (0.497)	-0.075 (0.570)
Focus Present	0.081 (0.304)	0.096 (0.105)	0.096 (0.101)	0.083 (0.157)
<i>Control variables</i>				
Firm Size	0.123 (0.767)	-0.186* (0.059)	-0.190* (0.053)	-0.163* (0.088)
ROA	-6.703 (0.102)	-7.551** (0.030)	-7.431** (0.030)	-7.468** (0.023)
Earnings Surprise	-0.302 (0.913)	0.090 (0.941)	0.228 (0.850)	0.225 (0.829)
Loss	-0.935** (0.036)	-0.224 (0.485)	-0.199 (0.541)	-0.217 (0.499)
MTB	-0.445** (0.014)	-0.196* (0.087)	-0.196* (0.089)	-0.164 (0.125)
Leverage	2.745** (0.028)	1.082* (0.055)	1.076* (0.055)	1.140** (0.039)
Following	0.259 (0.534)	0.192 (0.330)	0.193 (0.328)	0.169 (0.381)
Return	-2.201 (0.162)	-4.581*** (0.002)	-4.704*** (0.002)	-4.335*** (0.003)
Tenure	2.460***	0.402***	0.391***	0.373***

	(0.000)	(0.000)	(0.001)	(0.001)
Ownership	-97.103***	-35.230***	-35.082***	-34.647***
	(0.000)	(0.001)	(0.001)	(0.001)
Age	-6.579***	-4.109***	-4.002***	-3.752***
	(0.000)	(0.000)	(0.000)	(0.000)
Gender	0.527	0.666**	0.647**	0.618*
	(0.409)	(0.039)	(0.046)	(0.055)
Institutional Ownership	-0.338	-0.386	-0.364	-0.224
	(0.780)	(0.492)	(0.513)	(0.681)
Constant		10.930***	10.895***	9.887***
		(0.000)	(0.001)	(0.003)
Industry FEs		Yes	Yes	Yes
Year FEs		Yes		
Quarter FEs				Yes
Year-Quarter FEs	Yes		Yes	
Firm FEs	Yes			
Observations	3,766	32,696	32,692	32,696
Pseudo R ²	0.240	0.093	0.109	0.074